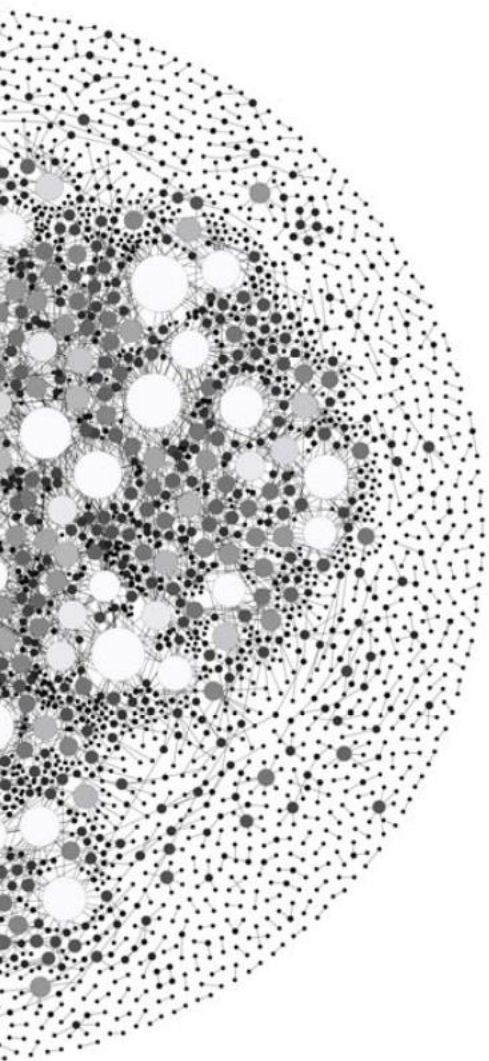




# Network estimation from time-series data I

Ria H. A. Hoekstra

January 29, 2026

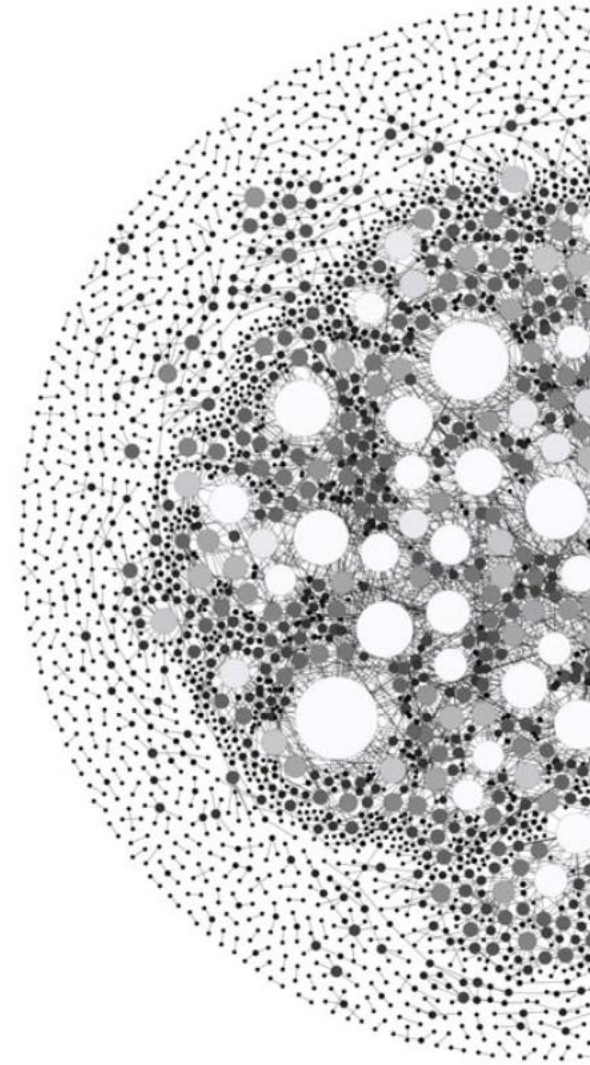
PSYNETS

University of Amsterdam

# This morning

---

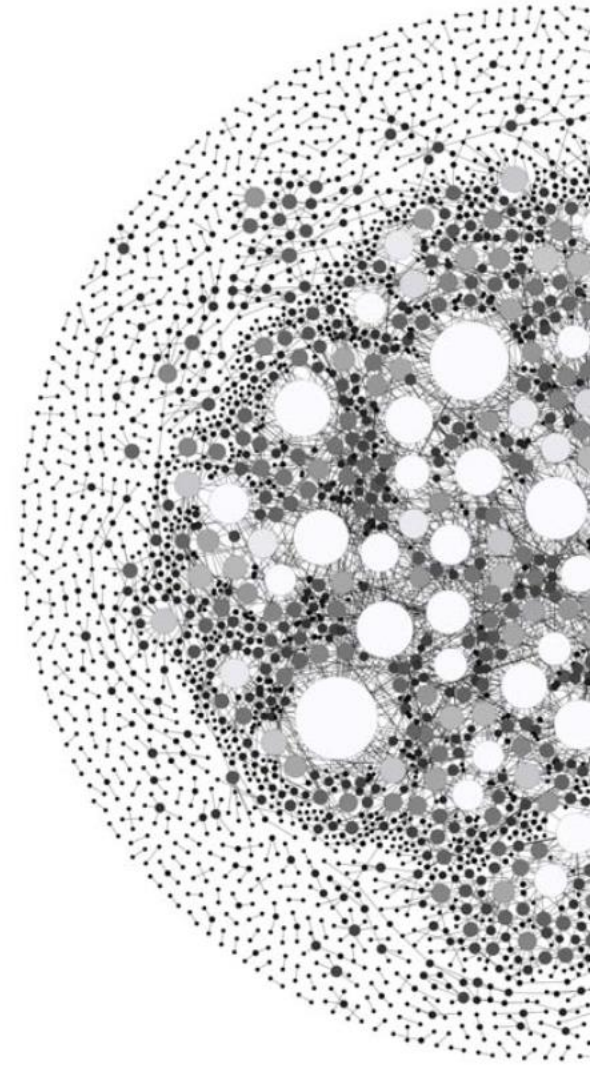
- Why model time series data?
- $N = 1$  &  $t = \text{large}$ 
  - Graphical Vector Autoregressive model (GVAR)
- Model assumptions
- Comparing  $N = 1$  network models
  - INIT



# This afternoon

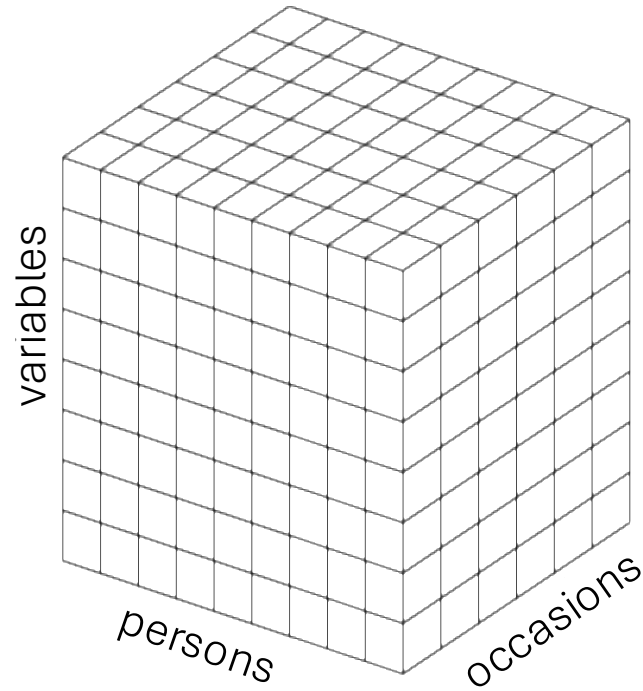
---

- $N > 1$  &  $t = \text{large}$ 
  - Multi-level VAR
- $N > 1$  &  $t = \text{small}$ 
  - Panel VAR
- $N = 1$  &  $t = \text{large}$ 
  - Mixed VAR
  - Time-Varying VAR



# Cross-sectional versus time series data

---

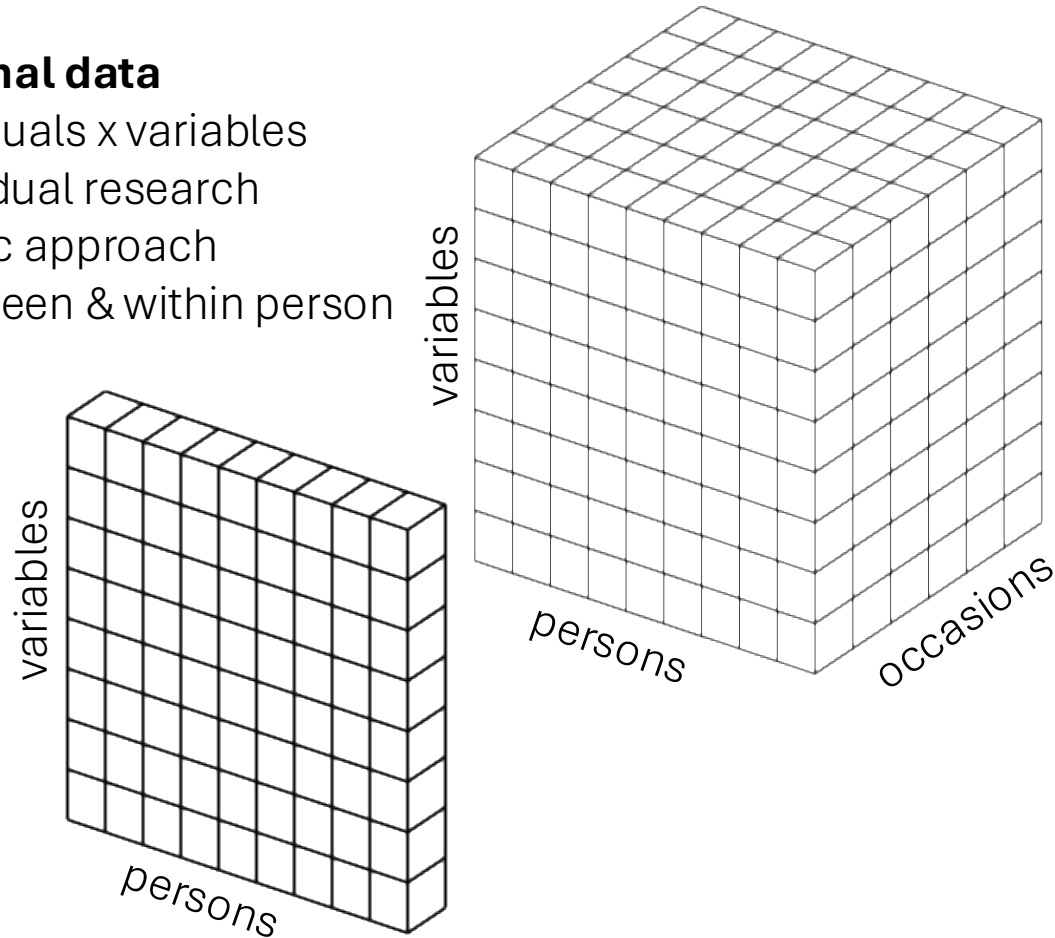


# Cross-sectional versus time series data

---

## cross-sectional data

- individuals x variables
- Inter-individual research
- Nomothetic approach
- Mix of between & within person variation

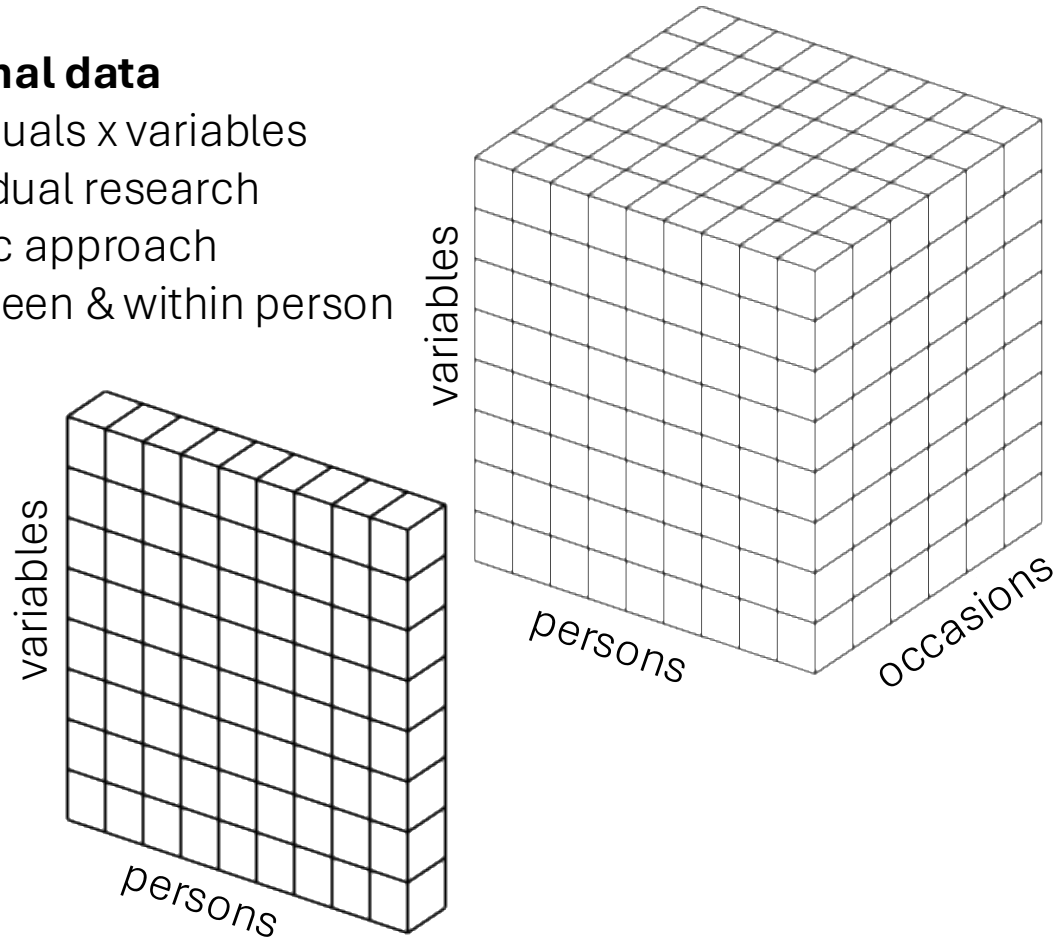


# Cross-sectional versus time series data

---

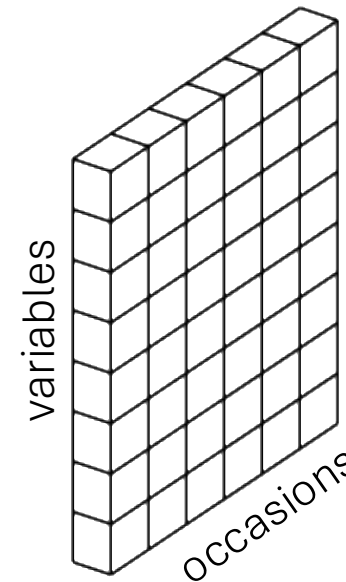
## cross-sectional data

- individuals x variables
- Inter-individual research
- Nomothetic approach
- Mix of between & within person variation



## time series data

- variables x occasions
- Intra-individual research
- Idiographic approach
- Within-person variation



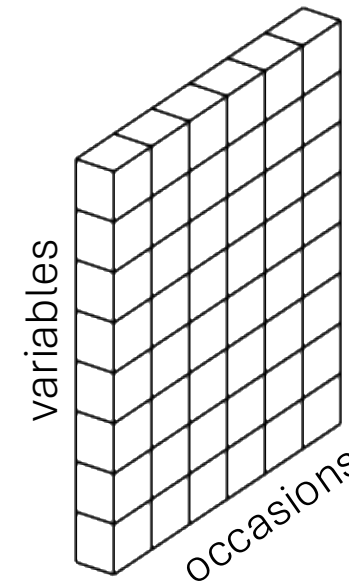
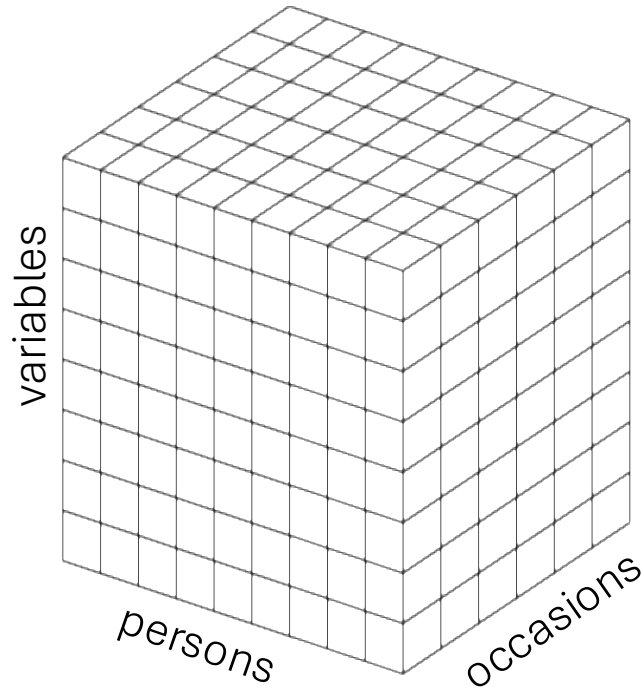
# Time series, ILD & panel data

## Intensive Longitudinal Data (ILD)

- variables x occasions  
( $t > 20$ ) x persons
- Mix of between & within person variation

## panel data

- variables x occasions  
( $t < 6$ ) x persons
- Mix of between & within person variation



## time series data

- variables x occasions
- Intra-individual research
- Idiographic approach
- Within-person variation

# What is time series data?

---

- Data collected over a series of time points for one or multiple individuals
- Different names for time series data
  - Daily Diary (DD)
  - Experience Sampling Method (ESM)
  - Ecological Momentary Assessment (EMA)
- Cases are **dependent**
  - There is a natural temporal ordering in the data
  - Observations closer together in time are more related than observations further apart in time





# Why model time series data?

---

- Gives insight into the **dynamic** relations between variables
  - Granger causality
- Difficult to translate cross-sectional results to individuals
- Cross-sectional results can only be translated directly to individuals when a system is **ergodic**:
  - Stationary
  - Homogenous

MEASUREMENT, 2(4), 201-218  
Copyright © 2004, Lawrence Erlbaum Associates, Inc.

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## FOCUS ARTICLE

---

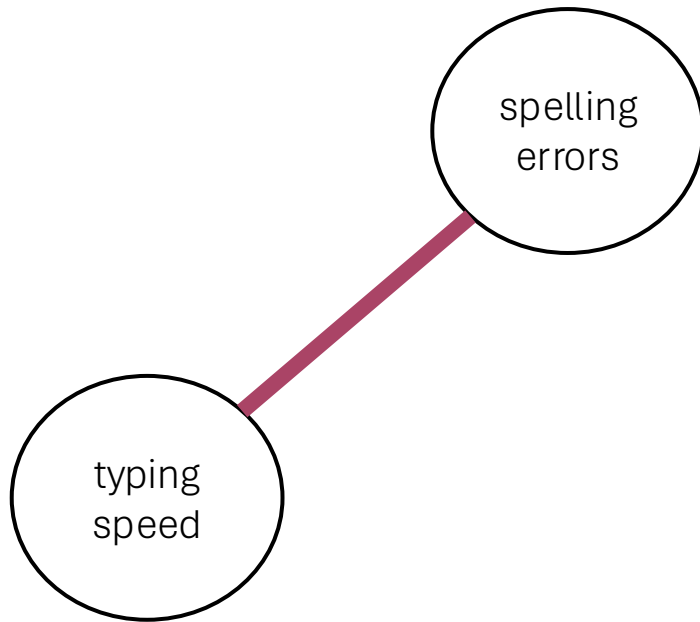
### A Manifesto on Psychology as Idiographic Science: Bringing the Person Back Into Scientific Psychology, This Time Forever

Peter C. M. Molenaar  
*Department of Psychology*  
*University of Amsterdam*

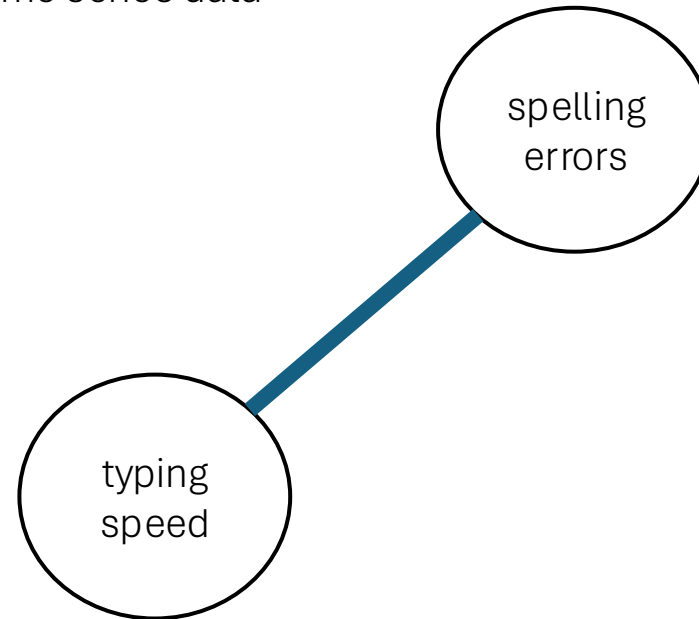
# Simpsons paradox

---

Cross-sectional data



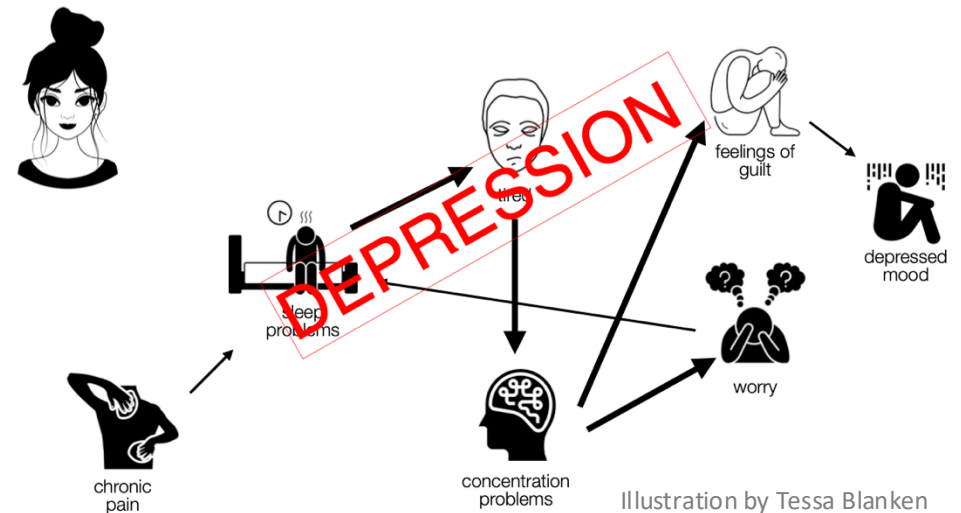
time series data

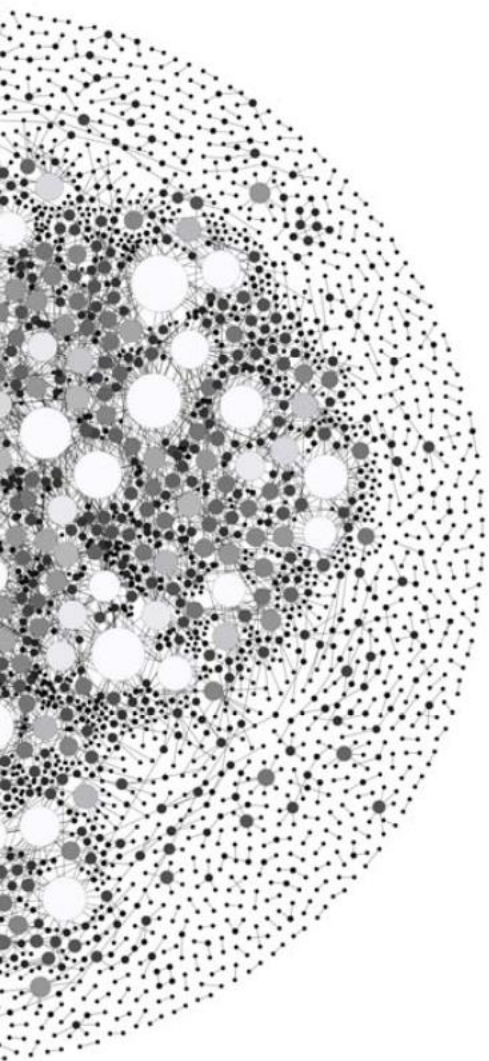


Example based on Hamaker (2012).

# Why model time series data as a network?

- Using time-series data, we can estimate network models for an individual
- This allows us to..
  - gain insight into individual processes
  - investigate individual differences
- Within the network approach to psychopathology, this is seen as a promising avenue to...
  - guide therapeutic interventions
  - personalize therapy



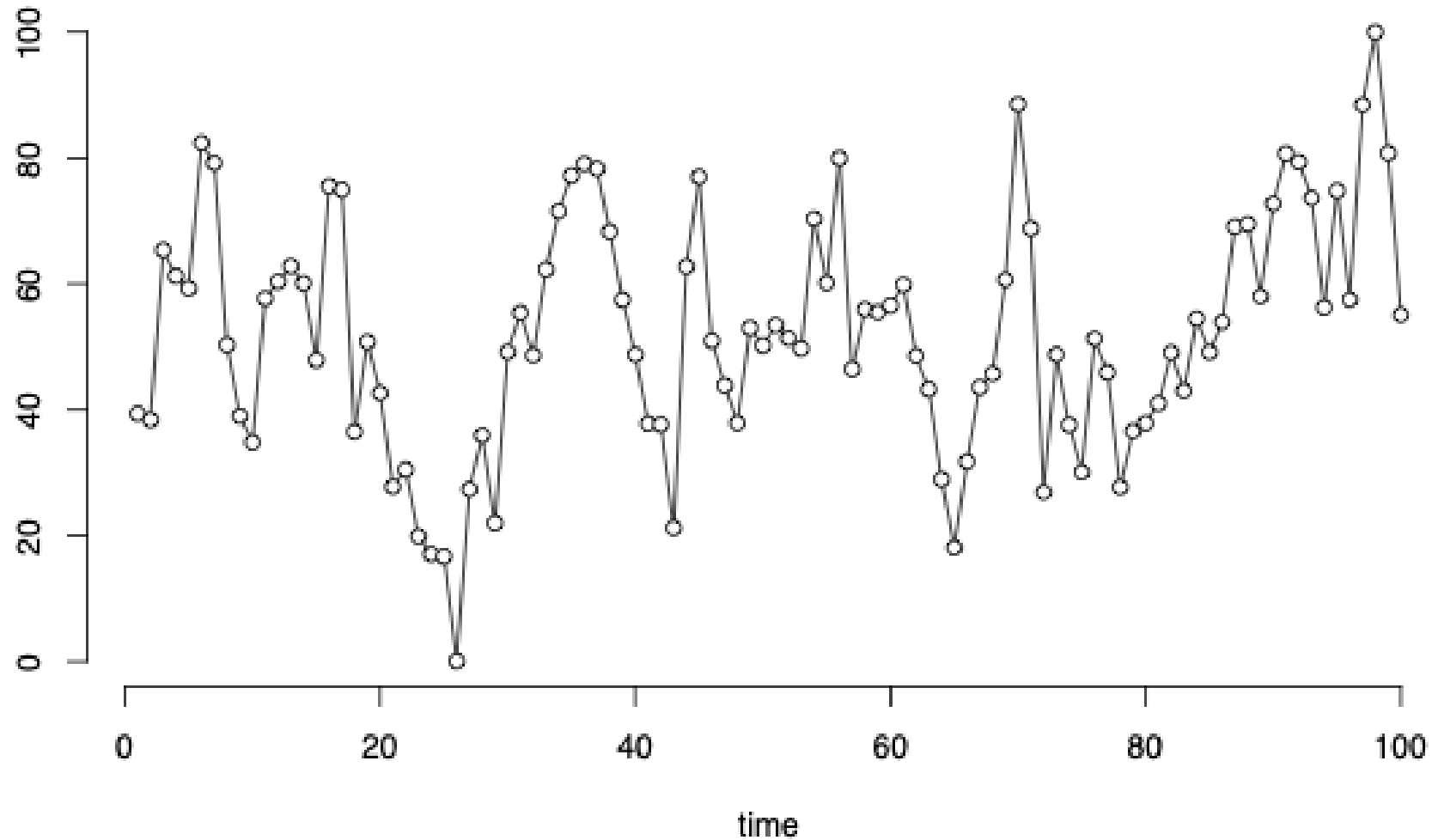


# N = 1 network

Modeling time-series dynamics for a single individual

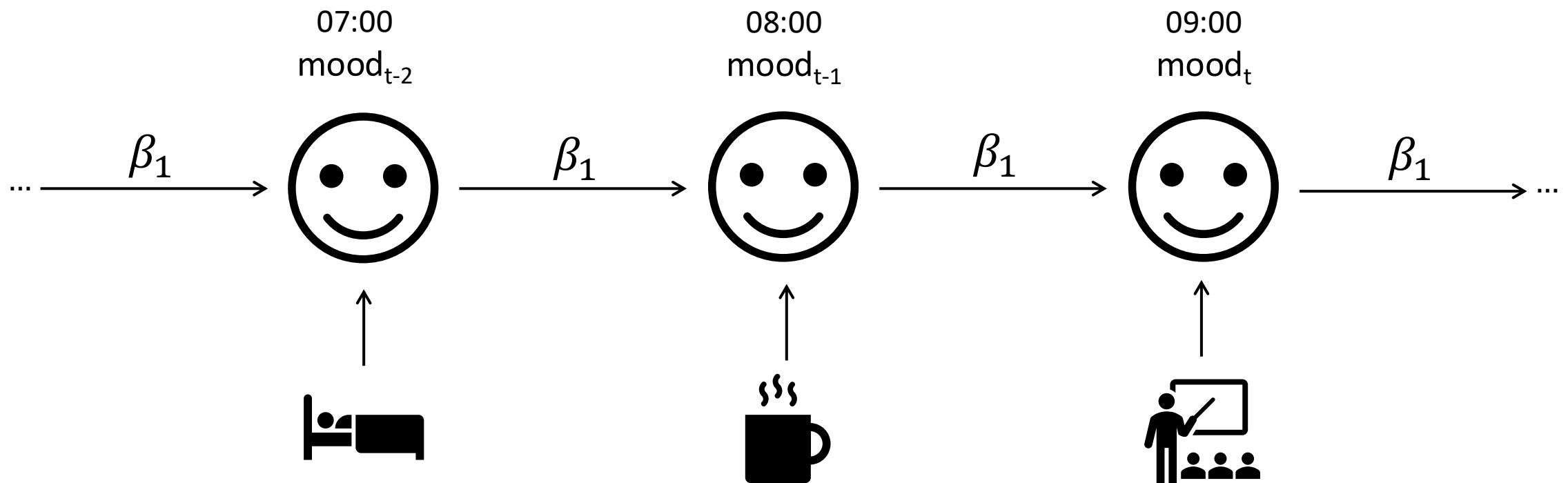
# Modeling **univariate** time series

---



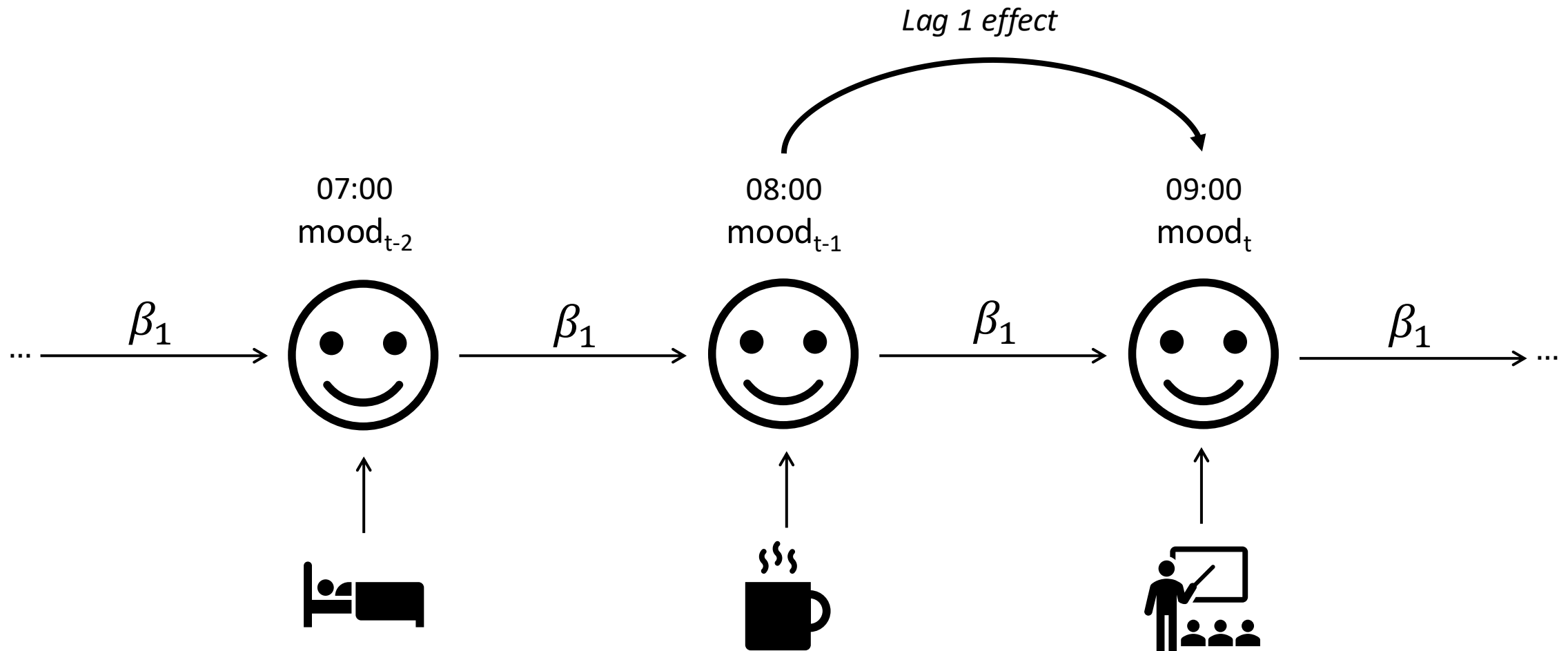
# Autoregressive (AR) Model

---



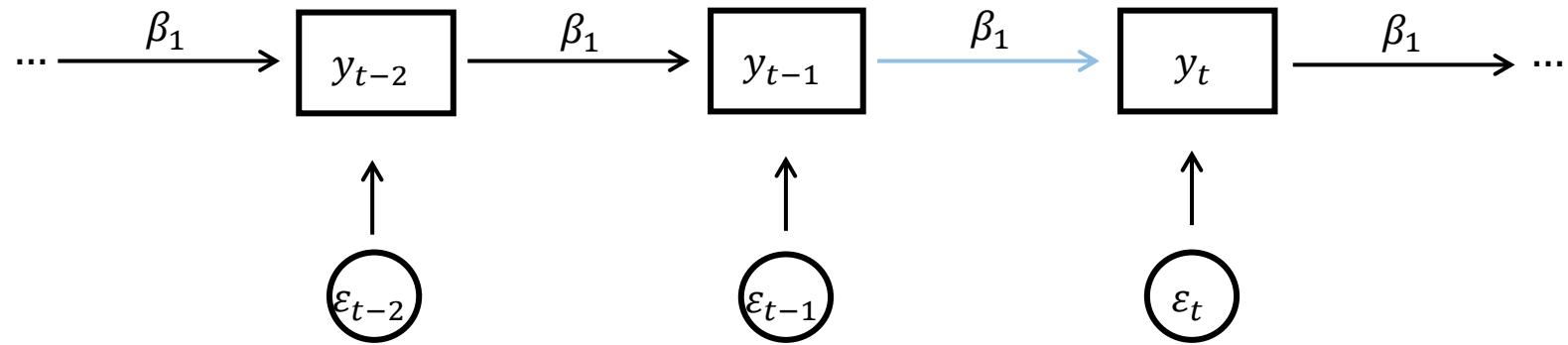
# Autoregressive Lag 1 Model: AR(1)

---



# AR (1) Model

---

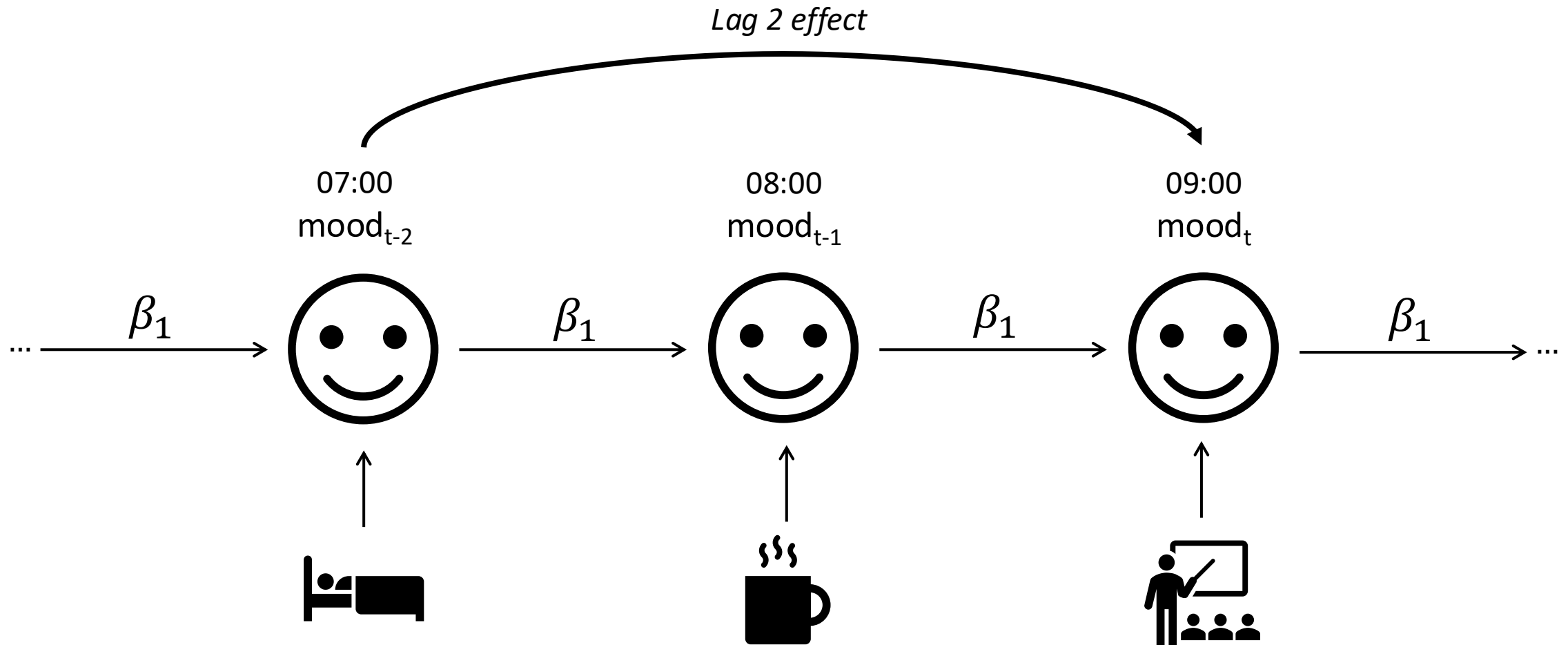


$$y_t = \boxed{\beta_1} y_{t-1} + \epsilon_t$$



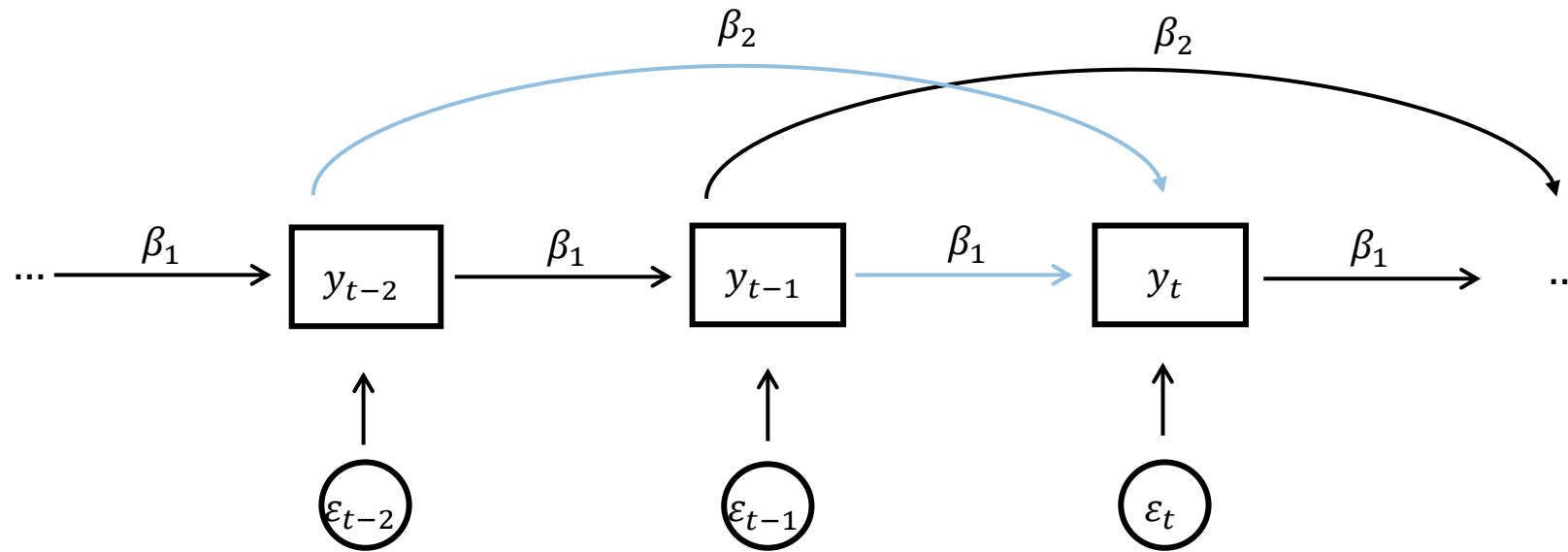
# Autoregressive Lag 2 Model: AR(2)

---



# AR (2) Model

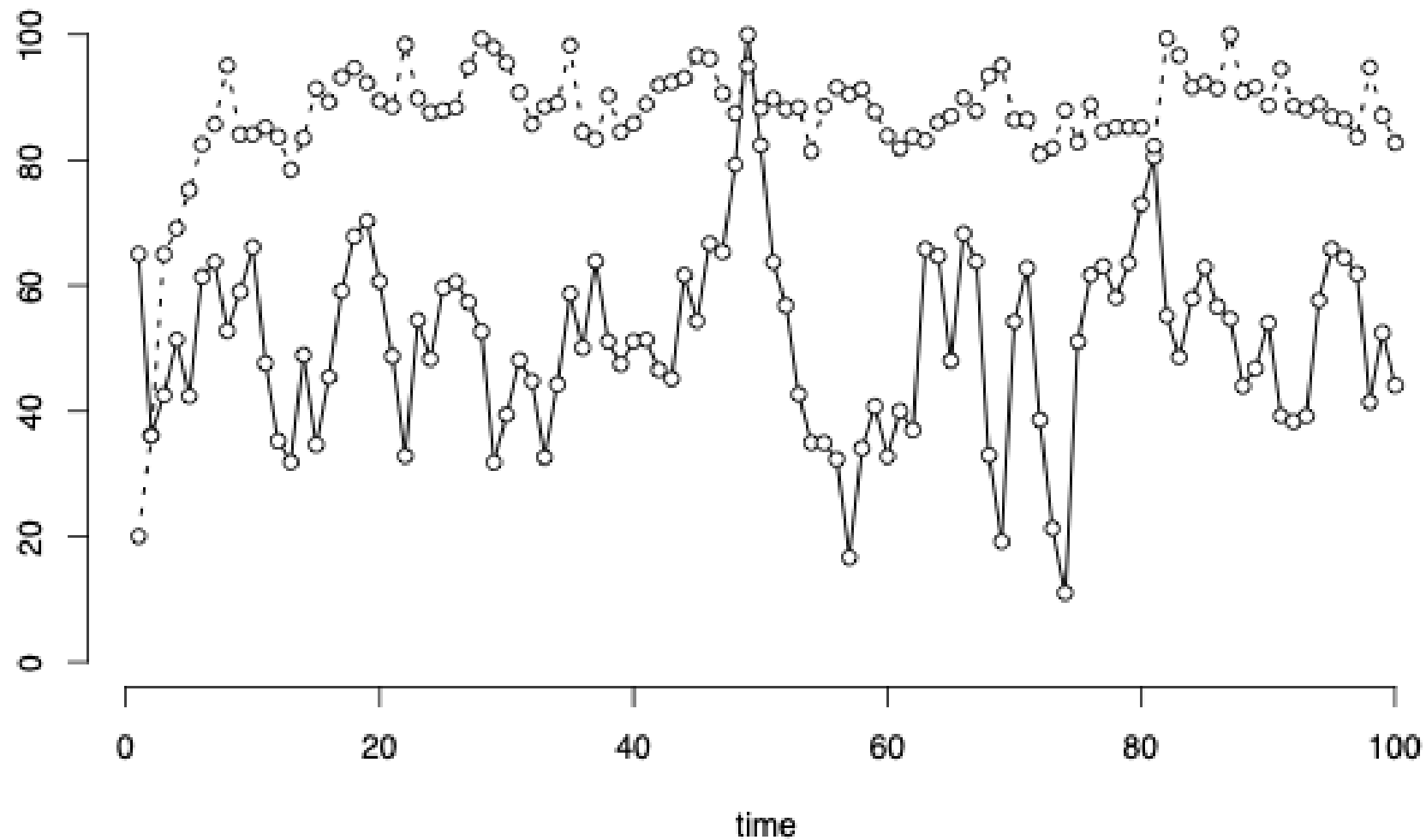
---



$$y_t = \beta_1 y_{t-1} + \beta_2 y_{t-2} + \varepsilon_t$$

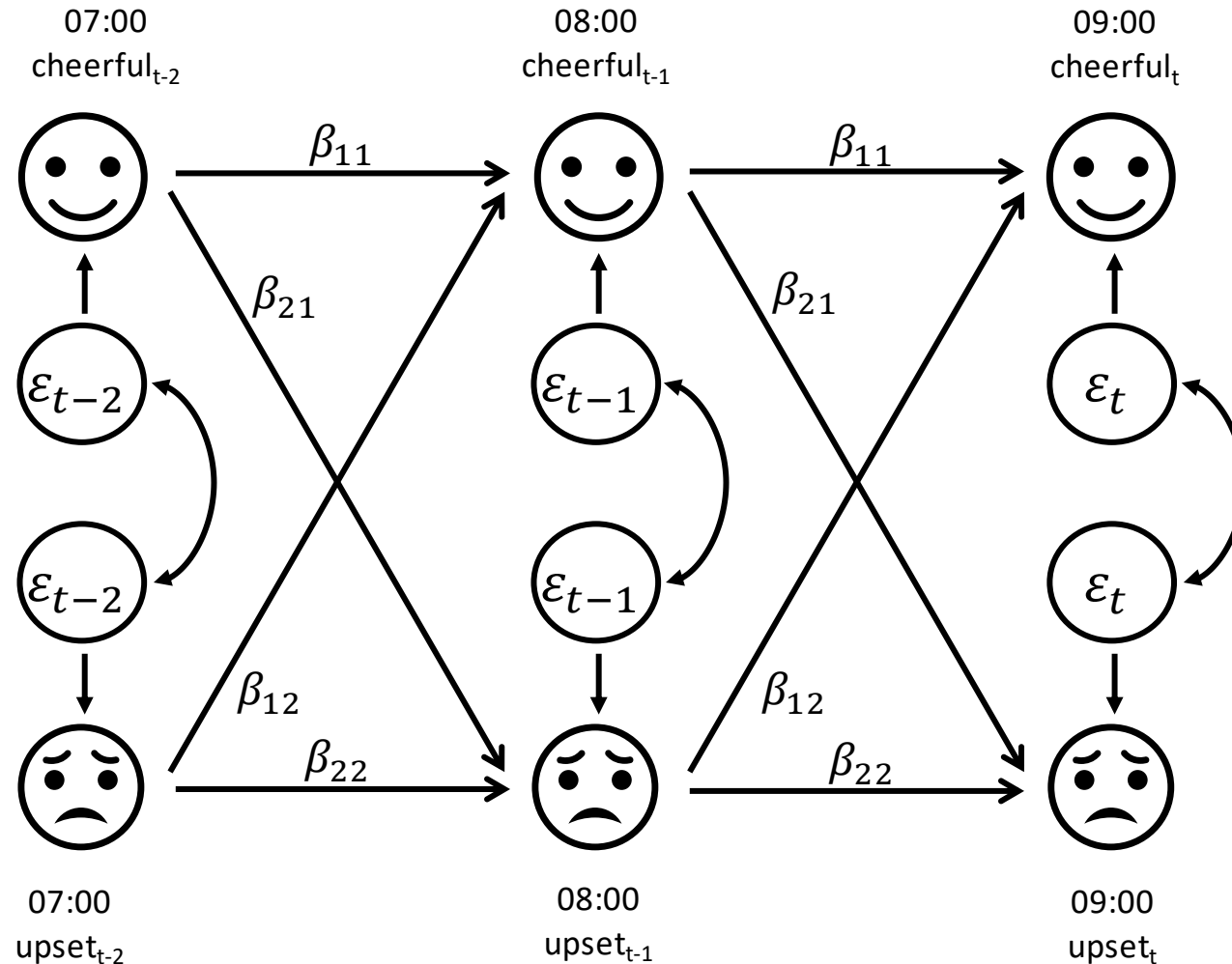
# Modeling **multivariate** time-series

---



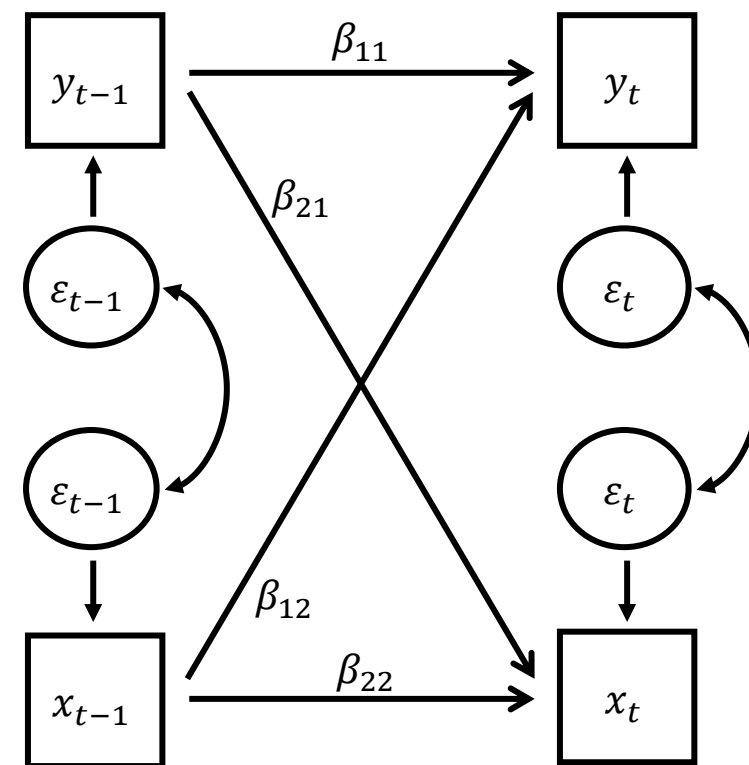
# Vector Autoregressive (VAR) model

---



# VAR model

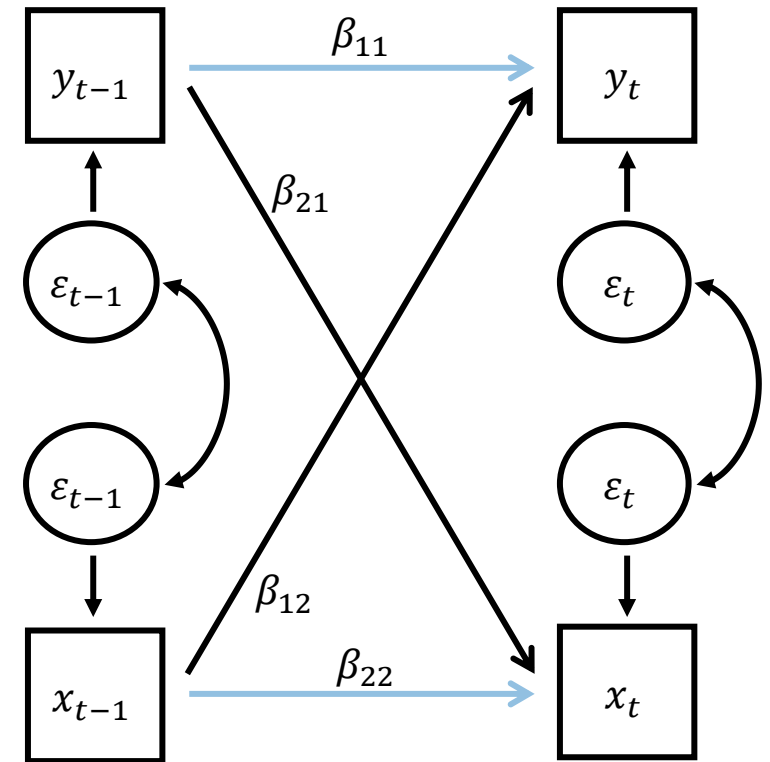
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# VAR model

---

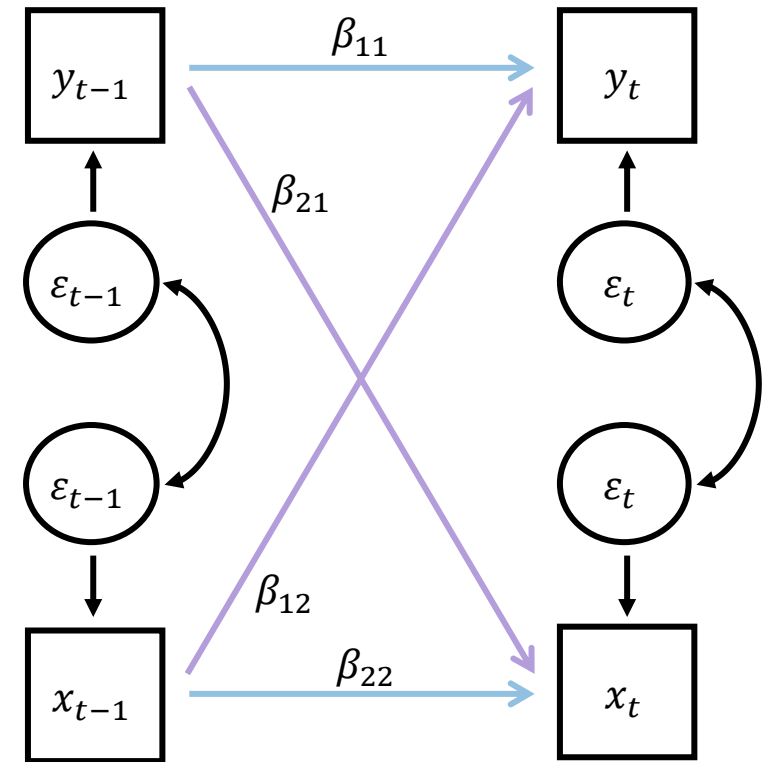
- Auto correlations
  - effect of one variable on itself



# VAR model

---

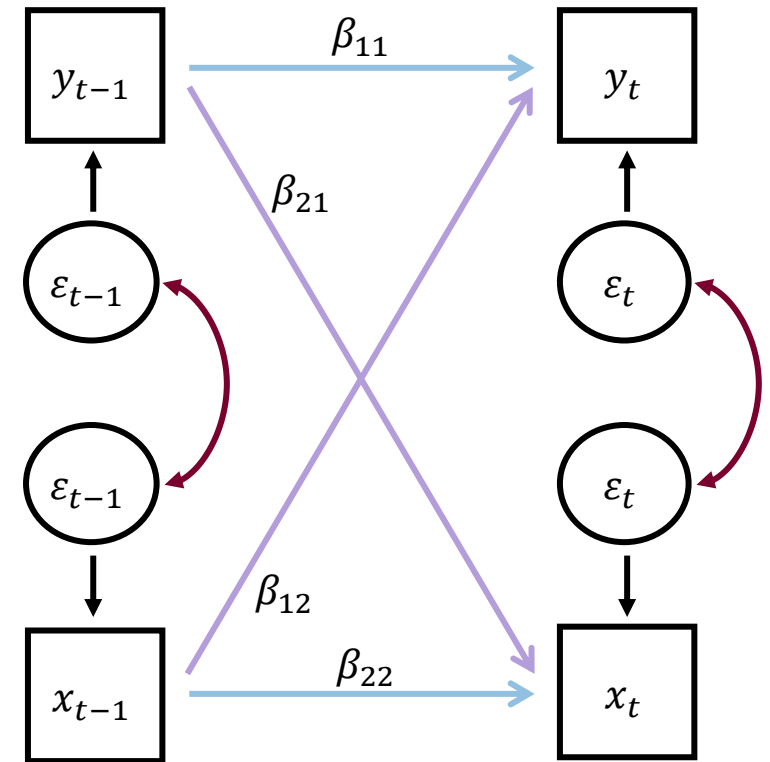
- Auto correlations
  - effect of one variable on itself
- Cross-lagged relations
  - effect of one variable on another variable



# VAR model

---

- Auto correlations
  - effect of one variable on itself
- Cross-lagged relations
  - effect of one variable on another variable
- Residuals
  - effect of variables on each other after taking time into account





# From VAR to Graphical VAR

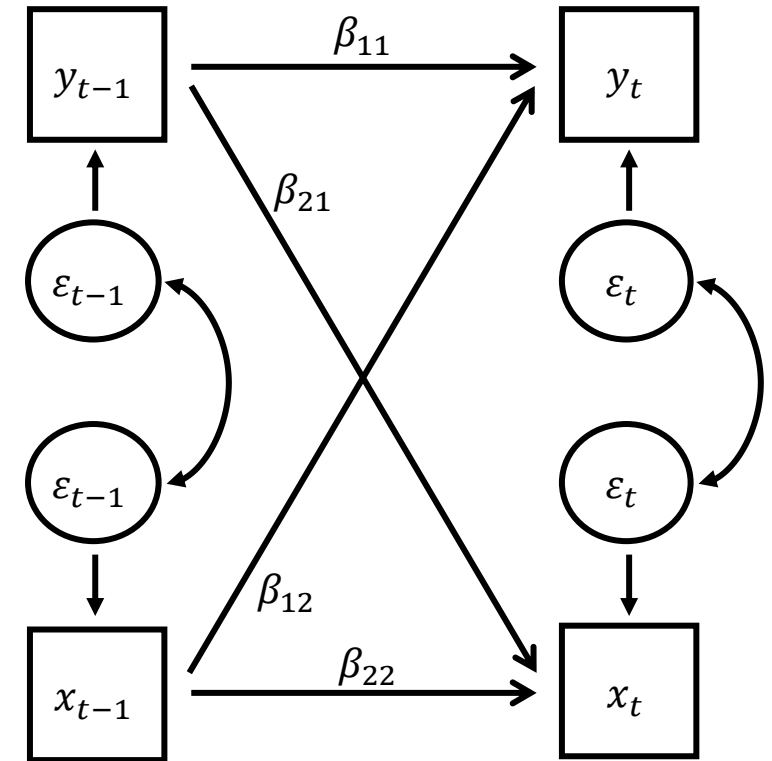
---

- **Temporal network**

- Shows if one variable predicts another variable in the next measurement occasion

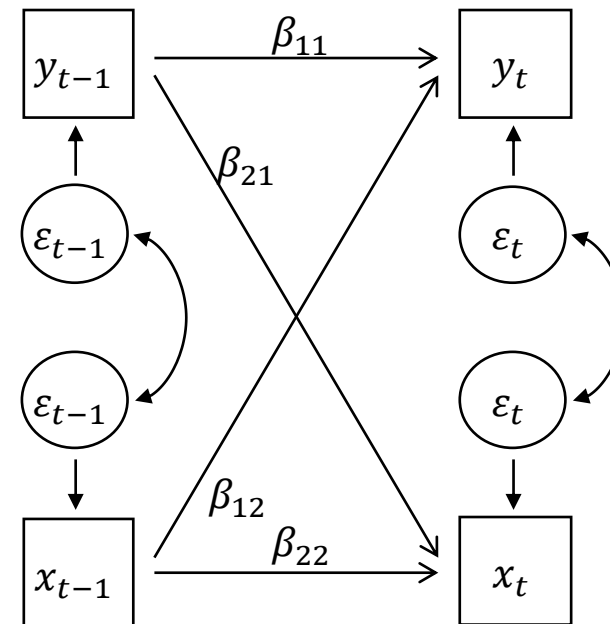
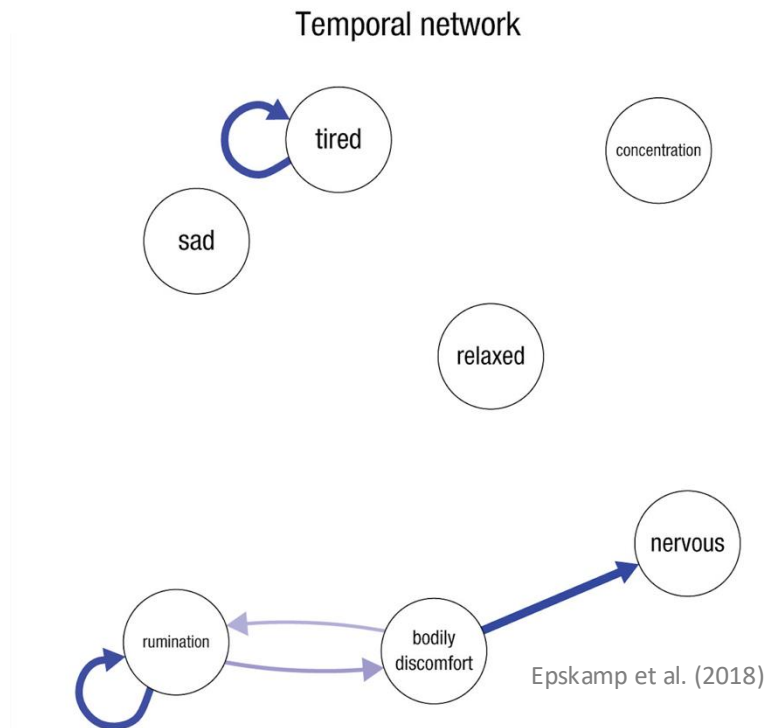
- **Contemporaneous network**

- Shows if two variables predict one another after taking temporal information into account



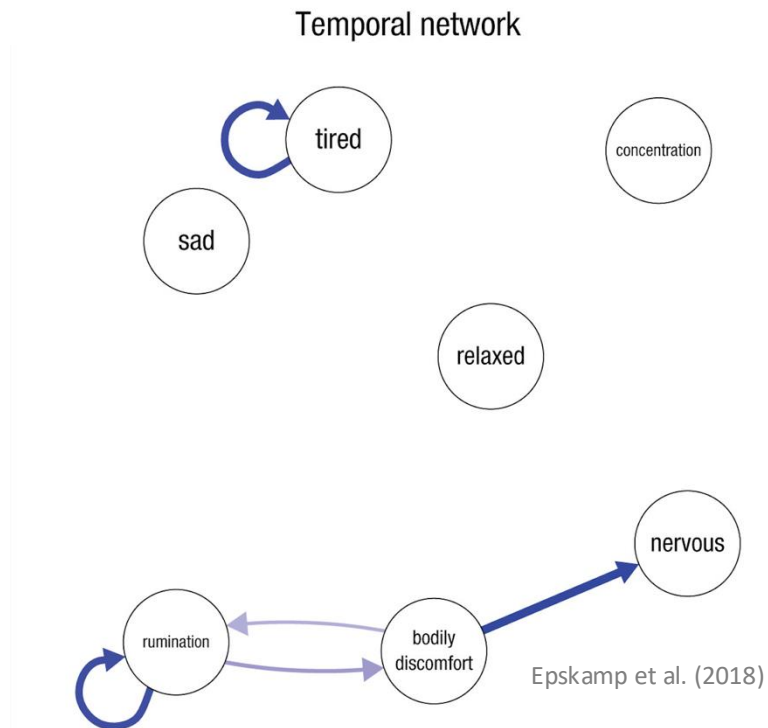
# From VAR to GVAR

- Temporal network
  - Shows that one variable predicts another variable in the next measurement occasion

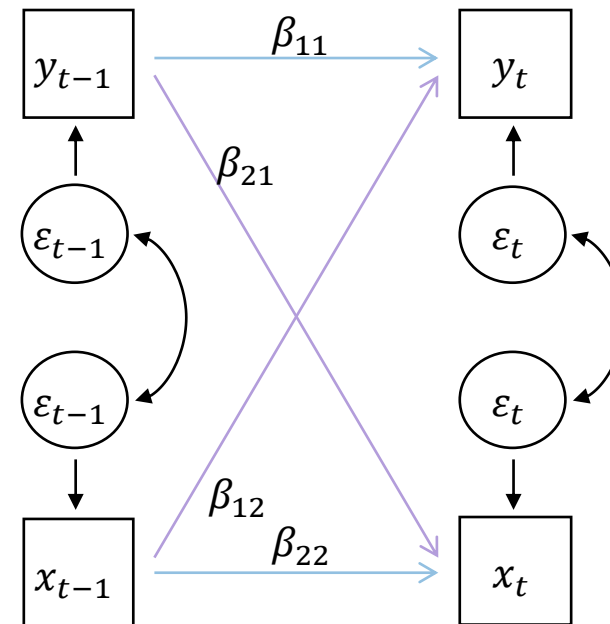


# From VAR to GVAR

- Temporal network
  - Shows that one variable predicts another variable in the next measurement occasion

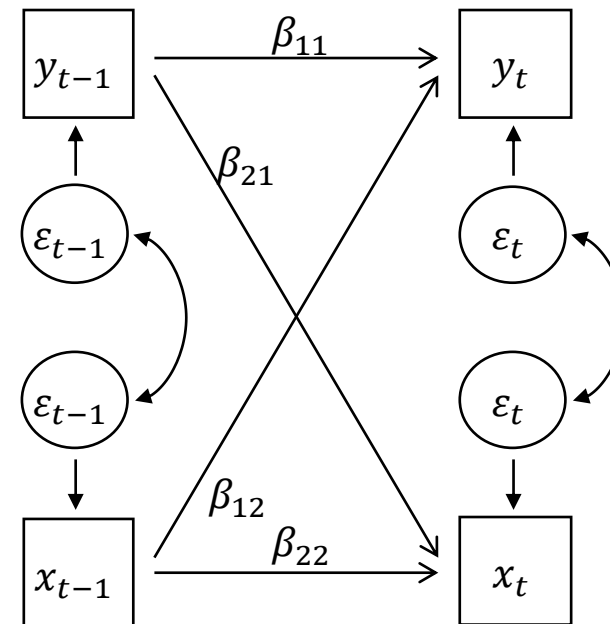
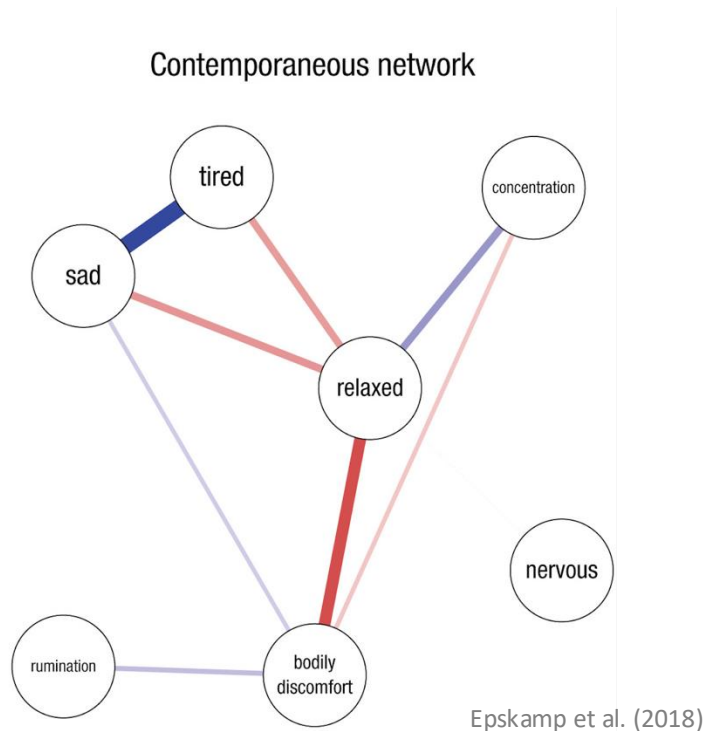


- Auto correlations
  - effect of one variable on itself
- Cross-lagged relations
  - effect of one variable on another variable



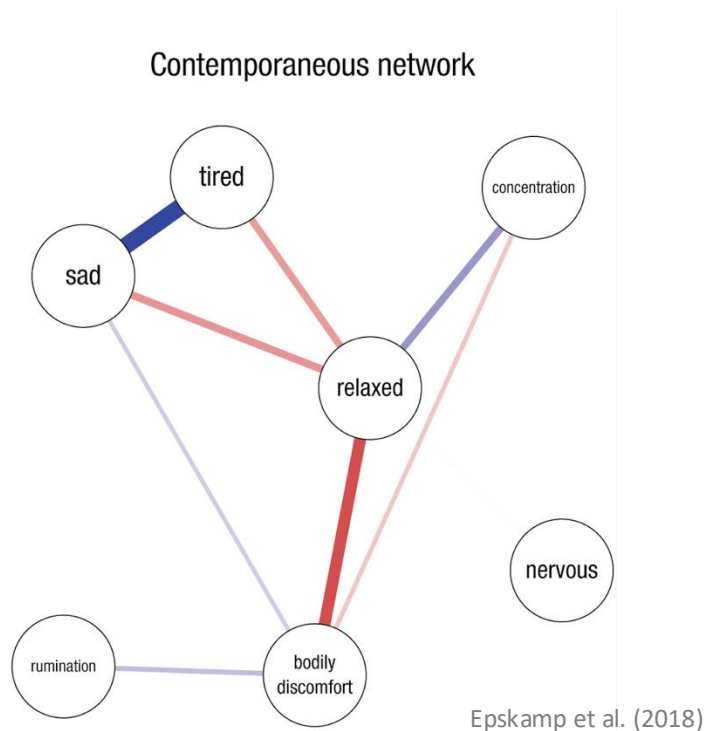
# From VAR to GVAR

- Contemporaneous network
  - Shows if two variables predict one another after taking temporal information into account

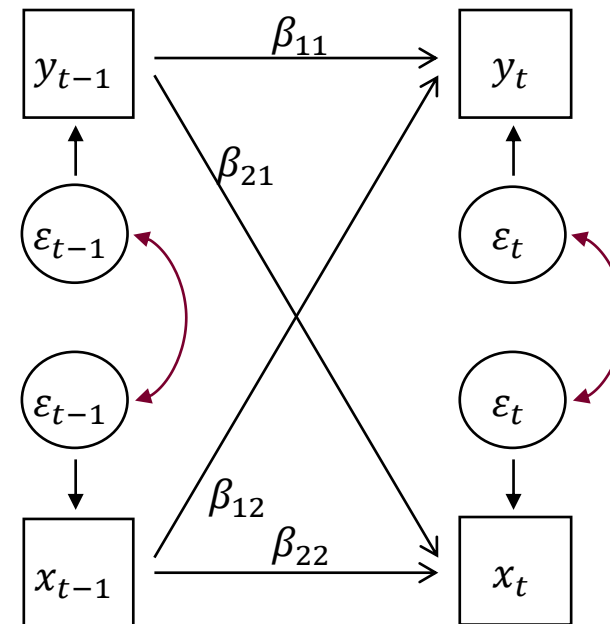


# From VAR to GVAR

- Contemporaneous network
  - Shows if two variables predict one another after taking temporal information into account



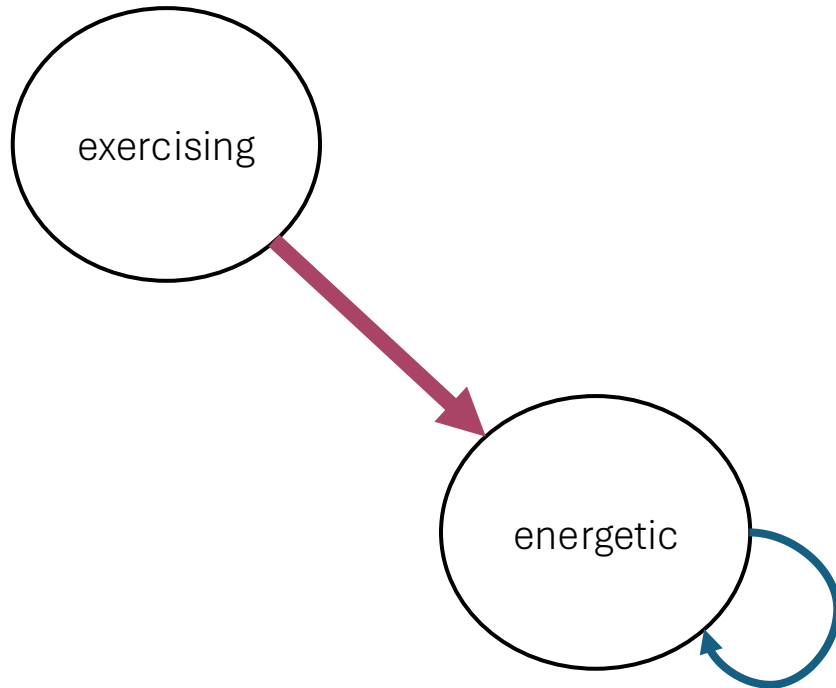
- Residuals
  - effect of variables on each other after taking time into account



# Temporal versus contemporaneous effects

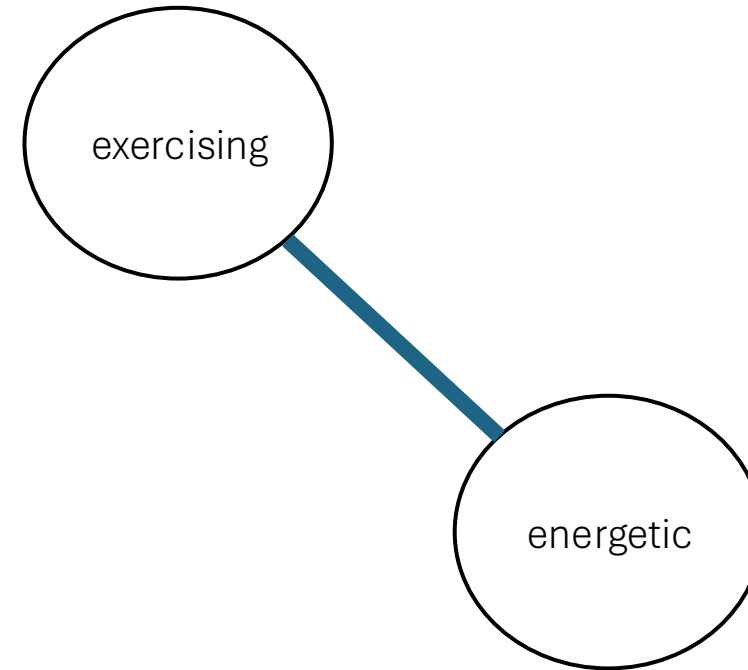
---

**Temporal network**



- Exercising now will lead me to be less energetic in the next time frame
- I am energetic now, and I will likely remain so in the next time frame

**Contemporaneous network**

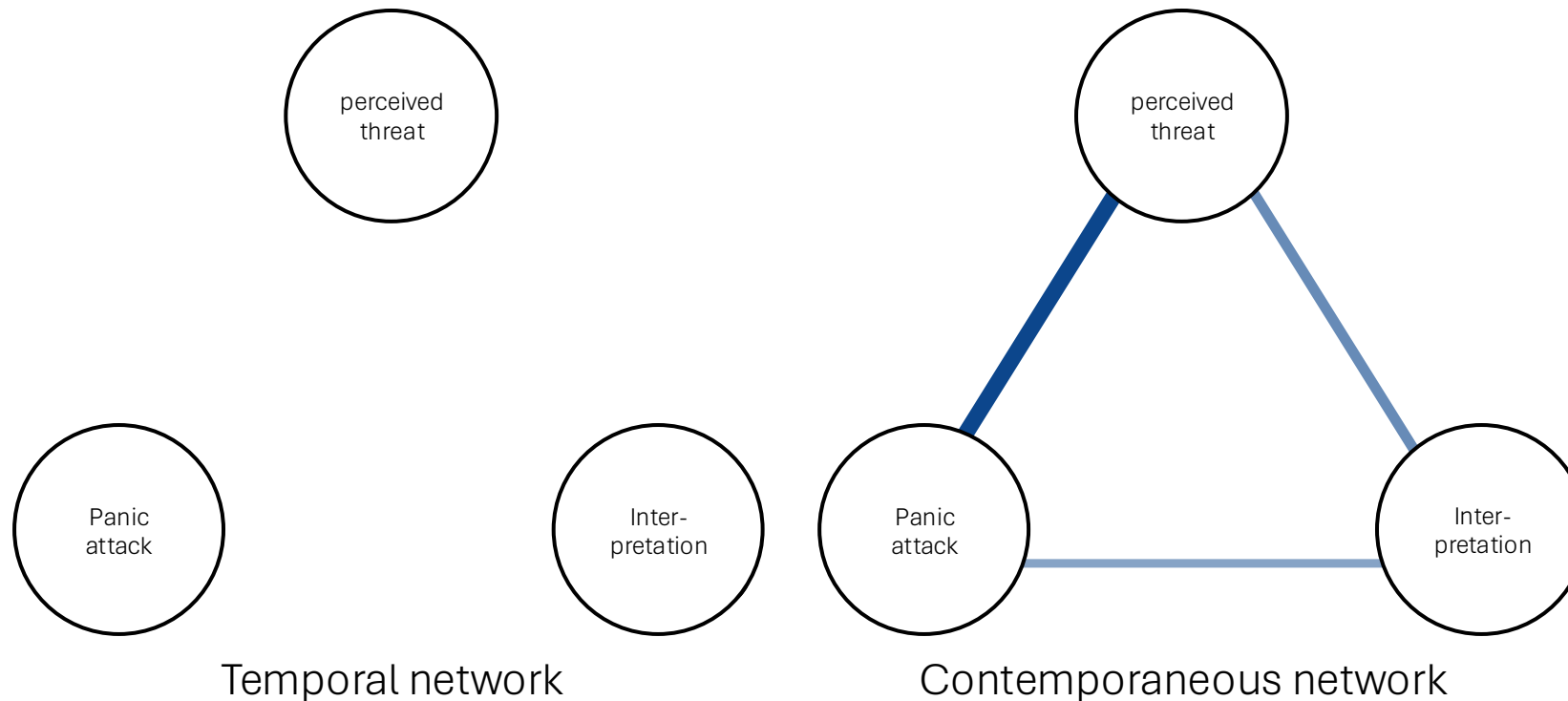


- Within the timeframe I am exercising, I feel energetic

# Contemporaneous network

---

- Shows if two variables predict one another after taking temporal information into account
- Contains effects **faster** than the time-window of measurement



# Graphical VAR model

---

- $\mathbf{y}$  represents a vector of responses for an individual at time  $t$
- $\mathbf{B}$  is a matrix containing the lagged regression coefficients
  - encodes the **temporal** network
- $\boldsymbol{\varepsilon}$  is a vector of residuals with a mean of zero and a time-invariant covariance matrix  $\boldsymbol{\Sigma}$
- $\mathbf{K} = \boldsymbol{\Sigma}^{-1}$ 
  - encodes the **contemporaneous** network
  - is modeled as a Gaussian Graphical Model (GGM)

$$\mathbf{y}_t = \mathbf{B}\mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_t$$

$$\boldsymbol{\varepsilon}_t \sim N(0, \boldsymbol{\Sigma})$$

$$\mathbf{K} = \boldsymbol{\Sigma}^{-1}$$



# Graphical VAR estimation

---

- Estimation **without** regularization/model selection:
  - Multiple regression (multivariate or univariate)
  - Stepwise non-regularized estimation implemented in the **psychonetrics** package

$$\mathbf{y}_t = \mathbf{B}\mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_t$$

$$\boldsymbol{\varepsilon}_t \sim N(0, \Sigma)$$

$$\mathbf{K} = \Sigma^{-1}$$

# Graphical VAR estimation with regularization

---

- Finding sparse solutions for  $\mathbf{B}$  and  $\mathbf{K}$  through regularization:
  - GLASSO estimation with EBIC model selection
    - Specify hyperparameter  $\gamma$
  - Implemented in **graphicalVAR**
  - Requires missing data handling beforehand through imputation techniques such as Kalman filter or weighted moving average

$$\mathbf{y}_t = \mathbf{B}\mathbf{y}_{t-1} + \boldsymbol{\varepsilon}_t$$

$$\boldsymbol{\varepsilon}_t \sim N(0, \Sigma)$$

$$\mathbf{K} = \Sigma^{-1}$$

# Graphical VAR estimation through model selection

---

- Finding sparse solutions for **B** and **K** through model selection:
  - First step: Stepwise non-regularized estimation
  - Second step: Multiple model selection options such as pruning or thresholding
  - Includes Full Information Maximum Likelihood (FIML) for missing data handling
  - Implemented in **psychonetrics**

psychonetrics

# Graphical VAR estimation

---

- After estimation, **B** and **K** can be standardized To obtain:
  - partial directed correlations (PDC)

$$PDC(y_i, y_j) = \frac{\beta_{ij}}{\sqrt{\Sigma_{ii}k_{jj} + \beta_{ij}^2}}$$

- partial contemporaneous correlations (PCC)

$$PCC(y_i, y_j) = -\frac{\kappa_{ij}}{\sqrt{\kappa_{ii}\kappa_{jj}}}$$

- These are the parameters we plot in the temporal network (PDC) and the contemporaneous network (PCC)

# graphicalVAR in R

---

- Main function:  
**graphicalVAR()**
- Important arguments:
  - **gamma**: EBIC hyper-parameter
  - **vars**: vector of variables to include in the analysis
  - **beepvar**: column name indicating assessment beep per day
  - **dayvar**: column name indicating assessment day

```
graphicalVAR (graphicalVAR) R Documentation
```

**Estimate the graphical VAR model.**

**Description**

Estimates the graphical VAR (Wild et al., 2010) model through LASSO estimation coupled with extended Bayesian information criterion for choosing the optimal tuning parameters. The estimation procedure is outlined by Rothman, Levina and Zhu (2010) and is further described by Abegaz and Wit (2013). The procedure here is based on the work done in the R package SparseTSCGM (Abegaz and Wit, 2014).

**Usage**

```
graphicalVAR(data, nLambda = 50, verbose = TRUE, gamma = 0.5, scale  
             = TRUE, lambda_beta, lambda_kappa,  
             regularize_mat_beta, regularize_mat_kappa, maxit.in =  
             100, maxit.out = 100, deleteMissings = TRUE,  
             penalize.diagonal = TRUE, lambda_min_kappa = 0.05,  
             lambda_min_beta = lambda_min_kappa, minic =  
             c("current", "0.1.2", "0.1.4", "0.1.5", "0.2"), vars,  
             beepvar, dayvar, idvar, lags = 1, centerWithin = TRUE,  
             likelihood = c("unnormalized", "normalized"))
```

# graphicalVAR in R

---

- Estimate model:
  - `res_gvar <- graphicalVAR(data, gamma = 0.5, vars = vars, dayvar = "day", beepvar = "beep")`
- Inspect temporal network:
  - `res_gvar$PDC`
- Inspect contemporaneous network:
  - `res_gvar$PCC`

# **gvar** in R (from psychometrics)

---

- Specify model:
  - `mod <- gvar(data, vars = vars, dayvar = "day", beepvar = "beep", estimator = "FIML")`
- Unregularized model estimation:
  - `mod <- mod %>% runmodel()`
- Pruning the model:
  - `mod <- mod %>% prune(alpha = 0.05) %>% runmodel()`

Note: different pruning and model selection options are possible, such as FDR-thresholding, and greedy set up search, and within each of these pruning options, you can make different choices again, such as correcting for multiple testing

# gvar in R (from psychonetrics)

---

- Evaluate model fit:
  - `mod %>% fit()`
- Estimated contemporaneous network:
  - `mod %>% getmatrix("omega_zeta")`
- Estimated temporal network:
  - `mod %>% getmatrix("PDC")`

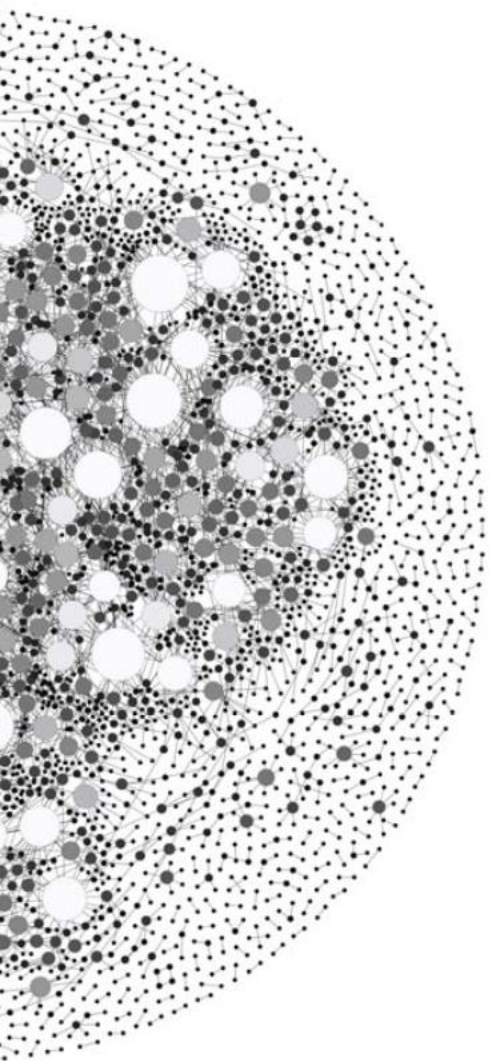
psychonetrics



# Conceptual questions

---

- True or False? If ergodicity holds, results from a cross-sectional analysis are expected to equal results from a within-person analysis.
- True or False? Edges in a contemporaneous network can be used to indicate effects that are faster and slower than the time window of measurement

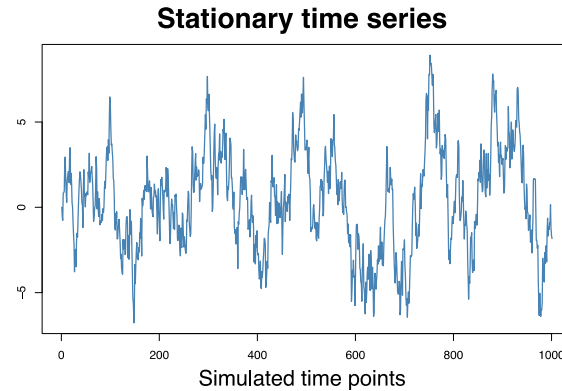


# Model assumptions & data challenges

# Assumptions of VAR: Stationarity

---

- Stationary time series means statistical properties such as mean, variance, autocorrelation, etc. are all **constant** over time
- Plausible when data is obtained within a short time span
- For a longer time span this might be unlikely to hold

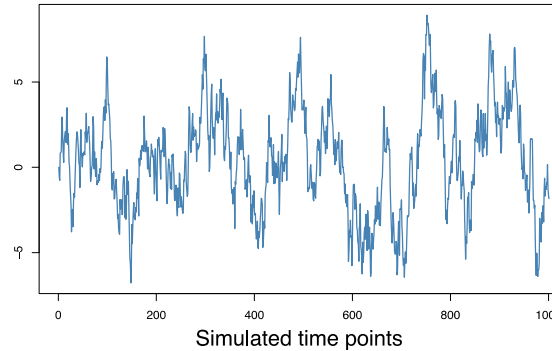


# Assumptions of VAR: Stationarity

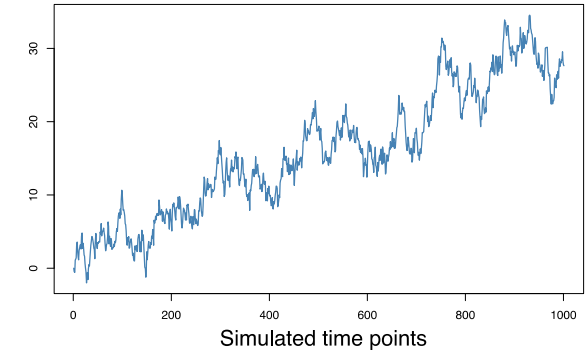
---

- Stationary time series means statistical properties such as mean, variance, autocorrelation, etc. are all **constant** over time
- Plausible when data is obtained within a short time span
- For a longer time span this might be unlikely to hold
- Different **trends** can be found in the data:
  - Deterministic trends
  - Stochastic trends

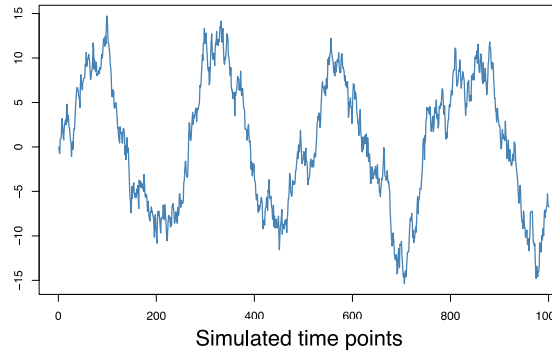
Stationary time series



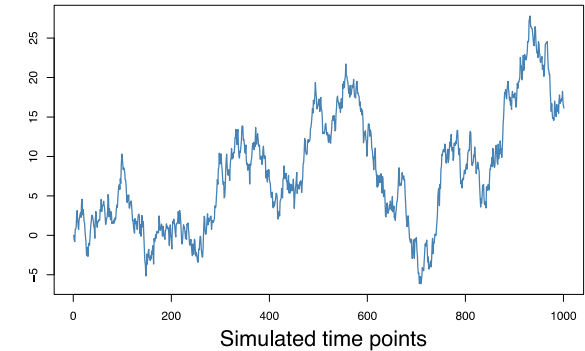
Non-stationary time series: **Linear trend**



Non-stationary time series: **Seasonal trend**



Non-stationary time series: **Unit root**



# Assumptions of VAR: Stationarity

---

- **Linear trends** can be assessed by regressing each time series on itself as a predictor
  - `kpss.test(data)` in R
    - $H_0$ : the time series is trend stationary
    - $H_A$ : the time series is not trend stationary
    - If  $p$  is smaller than the significance level (e.g.,  $\alpha = .05$ ) you reject the null and conclude the time series is not trend stationary
- Solutions:
  - **Detrending**: remove linear trends from data
    - However, often it is these trends that people are interested in
    - (linear) detrending can lead to induced trends in the data
  - Non-stationary network model estimation such as **time-varying VAR model**

# Detrending in R

---

- Check for linear trend in data
- Per person, regress each variable in your data (here “var1”) on itself using **time as a predictor**:

```
lm_var1 <- lm(var1 ~ time, data = data)
Summary(lm_var1)
```

- If significant, detrend the data by taking residuals as your new data

```
Data$var1[!is.na(data$var1)] <- residuals(lm_var1)
```

- Note: When estimating  $N = 1$  networks, you detrend each variable for each person separately

# Detrending in R

---

- Check for linear trend in data
- Per person, regress each variable in your data (here “var1”) on itself using **time as a predictor**:

```
lm_var1 <- lm(var1 ~ time, data = data)
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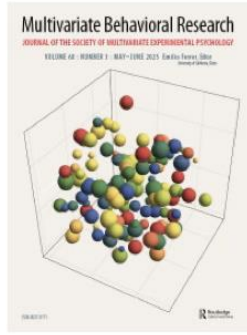
```
Data$var1[!is.na(data$var1)] <- residuals(lm_var1)
```

- Note: When estimating  $N = 1$  networks, you detrend each variable for each person separately

Detrending is not a magic trick: there can still be non-linear trends in your data or trends in your means, variances, etc!

# (Non)-Stationarity

---



## Multivariate Behavioral Research



ISSN: 0027-3171 (Print) 1532-7906 (Online) Journal homepage: [www.tandfonline.com/journals/hmbr20](http://www.tandfonline.com/journals/hmbr20)

## Non-Stationarity in Time-Series Analysis: Modeling Stochastic and Deterministic Trends

Oisín Ryan, Jonas M. B. Haslbeck & Lourens J. Waldorp

To cite this article: Oisín Ryan, Jonas M. B. Haslbeck & Lourens J. Waldorp (2025) Non-Stationarity in Time-Series Analysis: Modeling Stochastic and Deterministic Trends, Multivariate Behavioral Research, 60:3, 556-588, DOI: [10.1080/00273171.2024.2436413](https://doi.org/10.1080/00273171.2024.2436413)

To link to this article: <https://doi.org/10.1080/00273171.2024.2436413>



# Deterministic versus stochastic trends

 Utrecht University

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[About ILD](#)

[Your study](#) >  
[Theory](#) >  
[Measurement](#) >  
[Analysis](#) >

## Deterministic versus stochastic trends

AUTHORS  
Ellen L. Hamaker   
Ria H. A. Hoekstra 

AFFILIATION  
Methodology & Statistics Department, Utrecht University  
Methodology & Statistics Department, Utrecht University

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This article is about different kinds of trends over time that may be present in your data. You can think of a trend as a slow, long-term gradual movement in your data, that results in systematic change in a certain direction over time—typically upward or downward, although the direction can also change. While trends may not be of specific interest to you, you nevertheless have to account for them in your analysis: Failing to account for them may result in identifying dynamic patterns that are not reflective of the actual dynamics of the process, but rather of the long-term trend that characterizes the process.

[Analysis > Model overview >](#)  
Deterministic versus stochastic trends

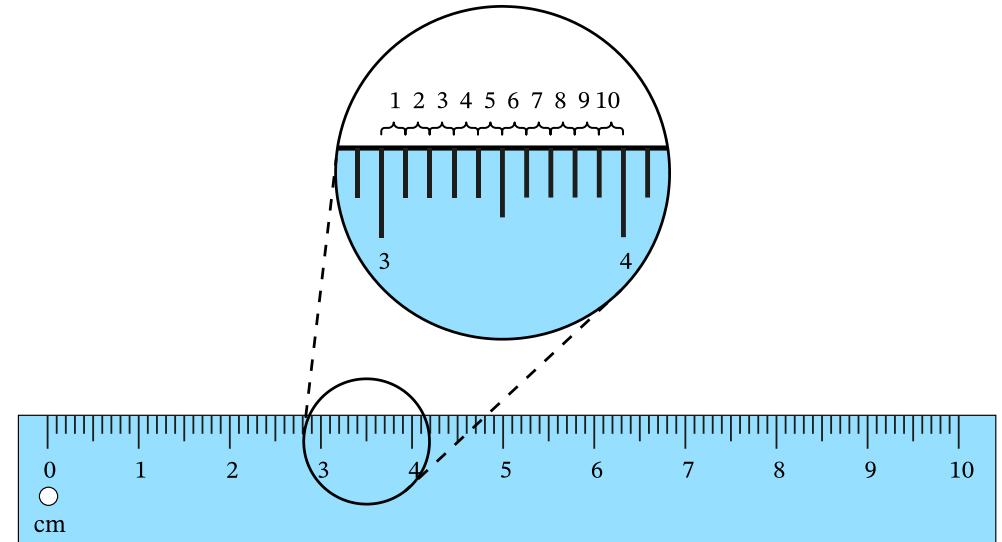
On this page

- 1 Some examples of a deterministic or stochastic trend
- 2 Detrending or differencing?
- 3 When a linear and stochastic trend are combined
- 4 How to detect a deterministic versus a stochastic trend
- 5 Think more about
- 6 Takeaway
- 7 Further reading

# Assumptions of VAR: Equidistant measures

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- When measuring over multiple days, this assumption is by definition violated due to the nights
- Possible solutions:
  - Create equal distances by adding missing data or aggregating data
  - Remove nights or model nights as missing observations (can be done in graphicalVAR and psychometrics package by specifying the day argument)
  - Continuous-Time (CT) VAR

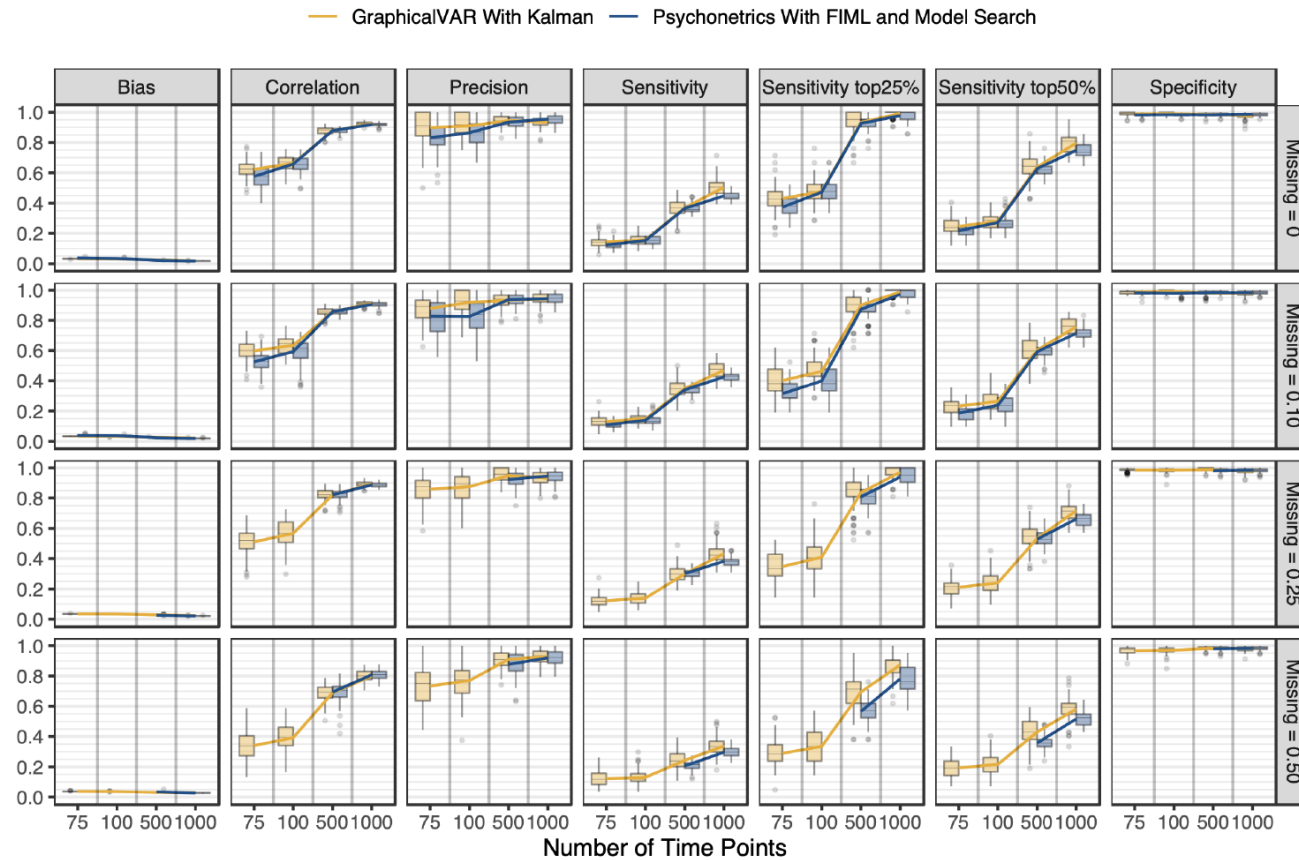


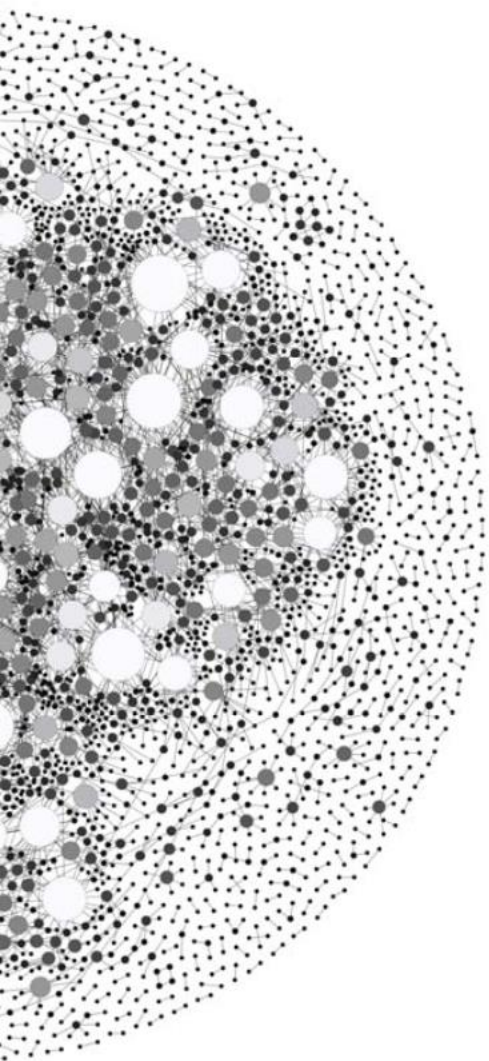
# Data challenge: Missing data

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- Often time-series data contains a lot of missingness
- Possible solutions:
  - Imputation
    - Many different imputation techniques possible
    - **Note:** most data imputation techniques assume the data is missing completely at random or at random
  - Estimating GVAR with FIML as implemented in psychometrics package
  - Deals with missing data through full maximum likelihood estimation

# Missing data: What works best?

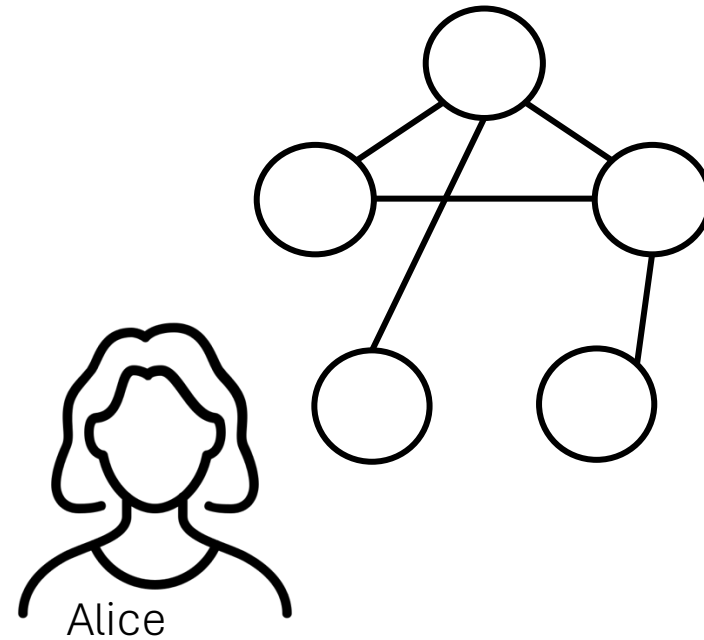
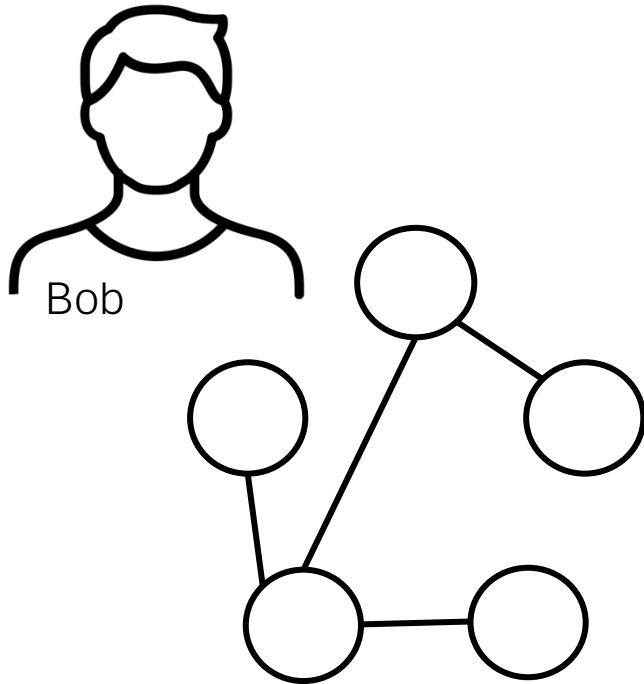




# Comparing $N=1$ networks

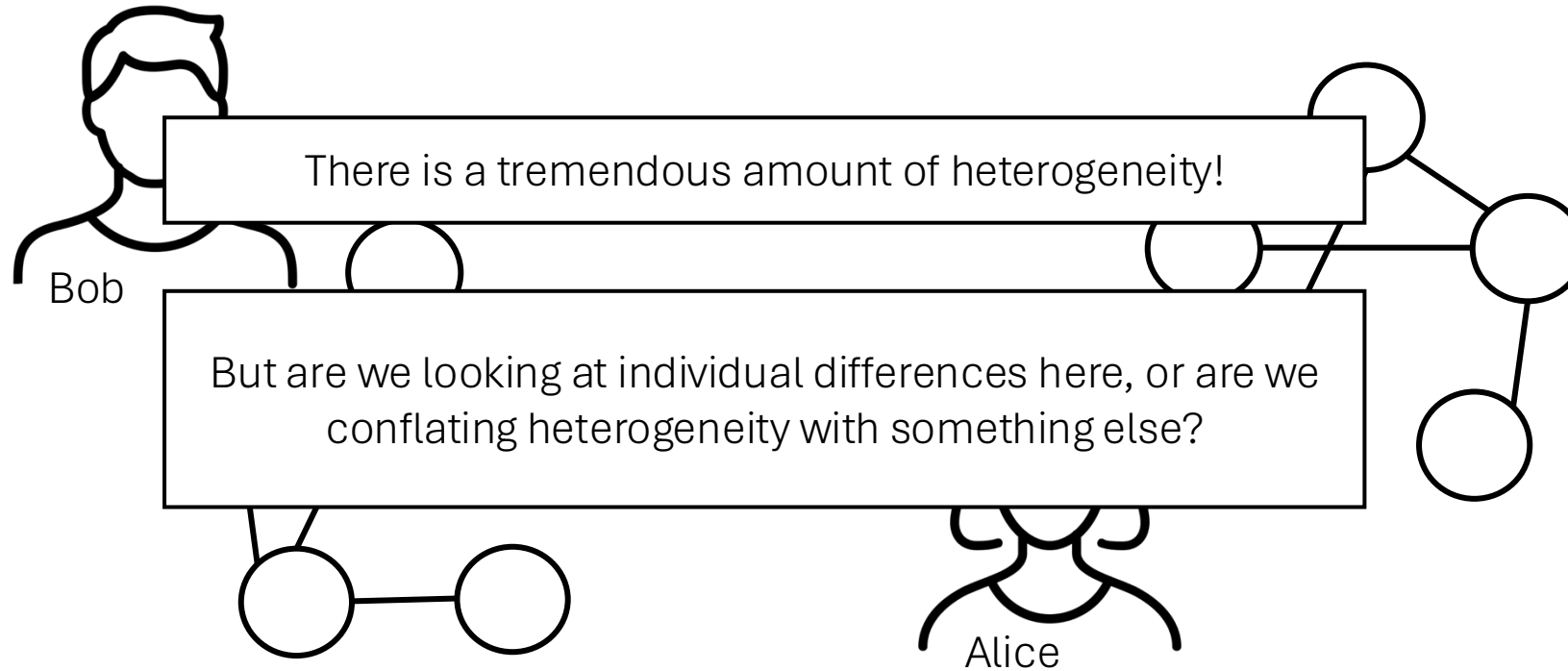
# Comparing idiographic networks

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# Comparing idiographic networks

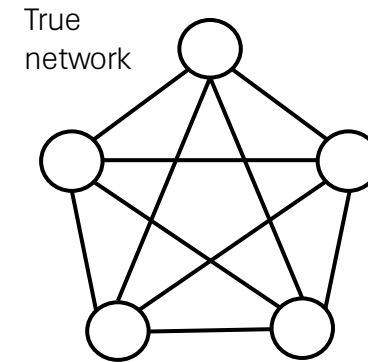
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# Heterogeneity: reality or illusion?

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- Suppose...
  - we measured 5 variables
  - the true network is fully connected
  - our sample is homogeneous
  - and we have a sensitivity of 50%



How much overlap do we expect to find if we randomly select three individuals from this homogeneous sample?

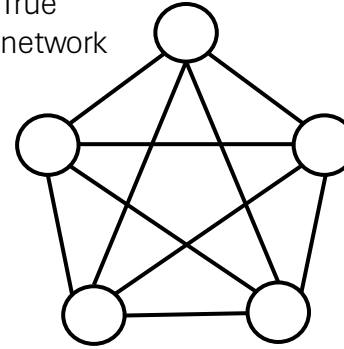


# Heterogeneity: reality or illusion?

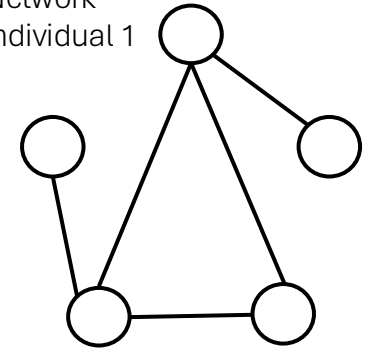
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- We expect to pick up 5 out of the 10 true edges for each individual
- On average, we would expect to find 25% of the edges to overlap between two randomly selected individuals
- The probability of obtaining the exact same network is less than 0.1% even though, in reality, all these individuals have the same network structure!

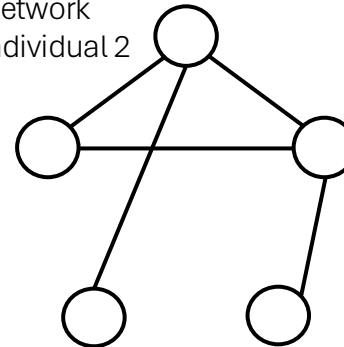
True network



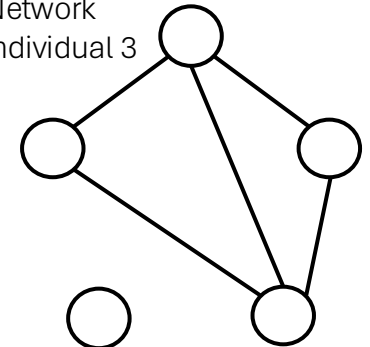
Network individual 1



Network individual 2



Network individual 3

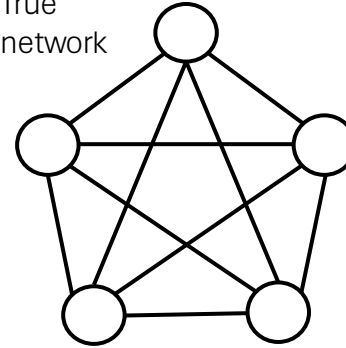


# Heterogeneity: reality or illusion?

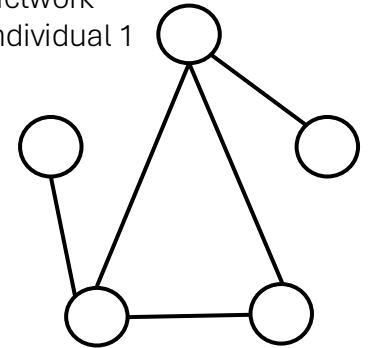
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- Conclusions:
  - Our estimation technique has a big influence on the amount of “heterogeneity” we find
  - Unreliable estimates result in the overestimation of heterogeneity
  - The comparison of idiographic network structures calls for a test

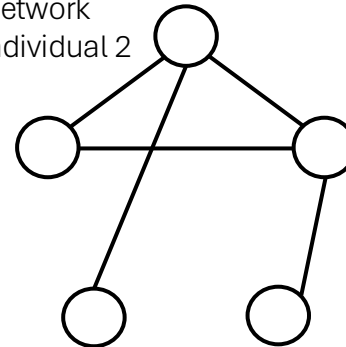
True network



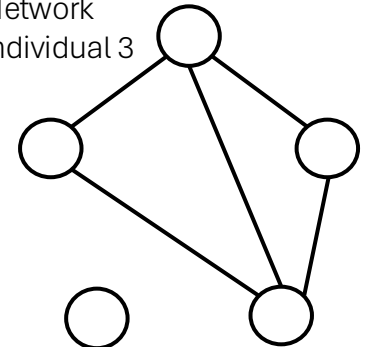
Network individual 1



Network individual 2

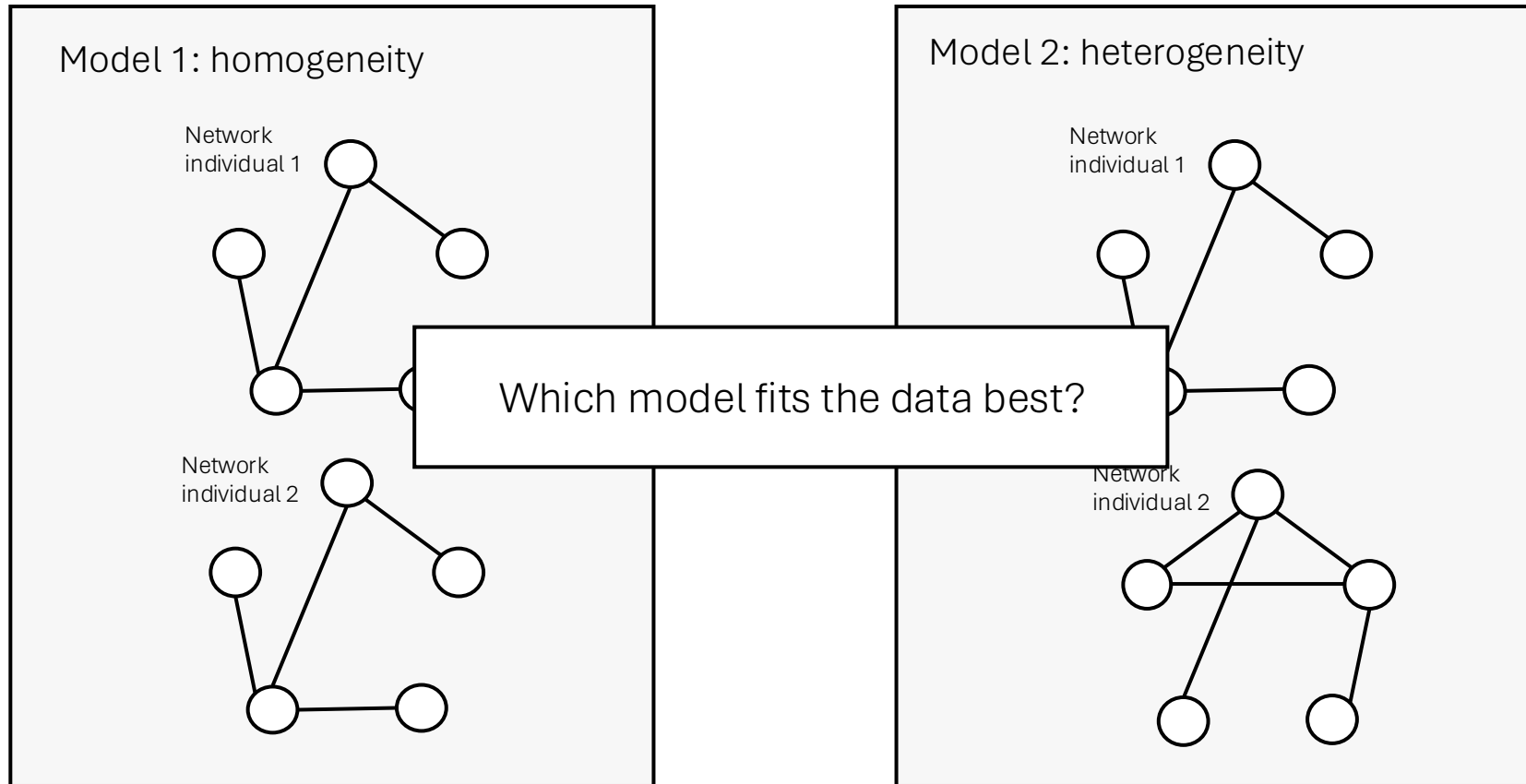


Network individual 3



# Individual Network Invariance Test (INIT)

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# Individual Network Invariance Test (INIT)

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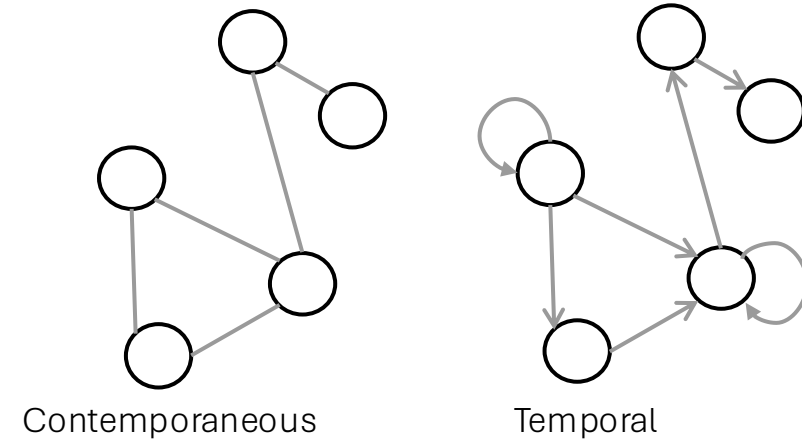
- The (lag-1) GVAR:

$$y_t = \beta y_{t-1} + \varepsilon_t,$$

$$\varepsilon_t \sim N(0, \Sigma)$$

- The variance-covariance structure ( $\Sigma$ ) can be modelled as a GGM:

$$\Sigma = K^{-1} = \Delta(I - \Omega)^{-1}\Delta$$



# Individual Network Invariance Test (INIT)

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- Individual Network Invariance Test (INIT) compares two models using equality constraints:

$$M_{\text{homogeneous}}: \Omega_1 = \Omega_2, B_1 = B_2$$

$$M_{\text{heterogeneous}}: \Omega_1 \neq \Omega_2, B_1 \neq B_2$$

- Using model comparison measures:
  - Akaike information criterion (AIC) when comparing **saturated** network models
  - Bayesian information criterion (BIC) when estimating **pruned** network models
- Implemented in the R package **INIT**

# INIT in R

---

- INIT requires you to specify:
  - **data**: the dataset containing data of both individuals
  - **idvar**: indicating the column name of the variable identifying the different individuals
  - **vars**: indicating the column names of the variables included in the network
- `INIT(data = data, idvar = colnames(data)[1], vars = colnames(data)[2:7])`

```
----- INIT results -----
```

```
The model without equality constraints is preferred.
```

```
INIT summary
```

- AIC homogeneity model: 6847.434
- AIC heterogeneity model: 6771.743
- delta AIC: -75.69085

```
input summary
```

- number of variables in the network: 6
- estimator used: ML
- estimated network type: saturated
- homogeneity test: overall

# INIT in R

---

- To **only** compare the temporal networks specifying the “**homogeneity\_test**” argument:
  - `INIT(data = data, idvar = colnames(data)[1], vars = colnames(data)[2:7], homogeneity_test = "temporal")`
- INIT makes use of the psychometrics package to estimate the model
- This means that the same arguments you can use in the psychometrics package to estimate a GVAR network can be specified when using **INIT**
  - `INIT(data = data, idvar = colnames(data)[1], vars = colnames(data)[2:7], estimator = "FIML", network_type = "pruned", homogeneity_test = "temporal")`



## Original Article

# Towards precision in the diagnostic profiling of patients: leveraging symptom dynamics as a clinical characterisation dimension in the assessment of major depressive disorder

Omid V. Ebrahimi, Denny Borsboom, Ria H. A. Hoekstra, Sacha Epskamp, Edoardo G. Ostinelli, Jojanneke A. Bastiaansen and Andrea Cipriani

**Background**

International guidelines present overall symptom severity as the key dimension for clinical characterisation of major depressive disorder (MDD). However, differences may reside within severity levels related to how symptoms interact in an individual patient, called symptom dynamics.

**Aims**

To investigate these individual differences by estimating the proportion of patients that display differences in their symptom dynamics while sharing the same overall symptom severity.

**Results**

Differential symptom dynamics (95% bootstrapped CI 41.0–8 severity group. The average difference was 2.2%.

**Conclusions**

The majority of participants in each symptom severity group displayed unique symptom dynamics. Examining symptom dynamics in person-specific psychopathology levels by revealing how sym

## RESEARCH ARTICLE

## Relating stability of individual dynamical networks to change in psychopathology

Sara van der Tuin<sup>1†\*</sup>, Ria H. A. Hoekstra<sup>2†</sup>, Sanne H. Booij<sup>1,3</sup>, Albertine J. Oldehinkel<sup>1</sup>, Klaas J. Wardenaar<sup>1</sup>, David van den Berg<sup>4,5</sup>, Denny Borsboom<sup>2</sup>, Johanna T. W. Wigman<sup>1</sup>

**1** University Medical Center Groningen, Department of Psychiatry, Interdisciplinary Centre Psychopathology and Emotion regulation, University of Groningen, Groningen, The Netherlands, **2** Department of Psychological Methods, University of Amsterdam, Amsterdam, The Netherlands, **3** Center for Integrative Psychiatry, Lentis, Groningen, The Netherlands, **4** Department of Clinical Psychology, Amsterdam Public Health Research Institute, Vrije Universiteit, Amsterdam, The Netherlands, **5** Department of Psychosis Research, Parnassia Psychiatric Institute, The Hague, The Netherlands

† ST and RHAH shared first authorship on this work.

\* [s.van.der.tuin@umcg.nl](mailto:s.van.der.tuin@umcg.nl)



# Discussion

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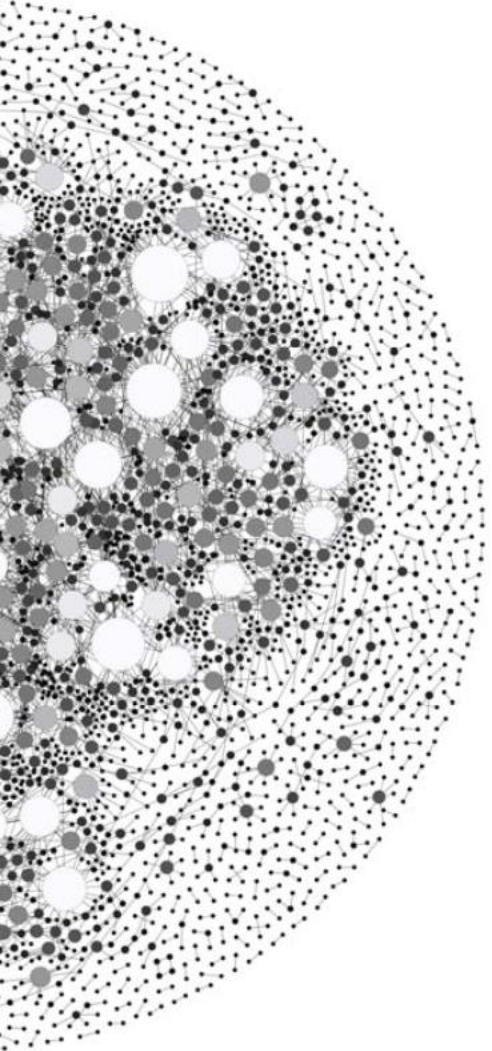
- Are there examples within your own field where time-series data might provide better insights than cross-sectional data? Why does temporal ordering matter in that context?
- Is the assumption of stationarity a reasonable assumption for the data you encounter in your field?

# Wrap up

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- Time series data allows us to study dynamics within an individual
- $N = 1$  &  $t = \text{large}$ 
  - Graphical VAR model
    - Assumes stationarity
    - Assumes equidistant measures
    - Missing data, as well as a lack of variation, is a challenge
- We can compare  $N = 1$  networks by placing equality constraints on the estimated network structures

Questions?



# Practical

You can work on the assignment